claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_7049e1ed-b93\agent-a82ce35ba77fa5f15.jsonl`
- SHA-256: `090f6e52115e004ca165fa308418dafde553dcd9d2d3dff8e4f32794467ba167`
- Source modified: `2026-07-02T04:58:36+00:00`
- Imported at: `2026-07-05T16:48:24+00:00`
- project: `wf_7049e1ed-b93`
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`

Transcript

1. user (2026-07-02T04:45:39.424Z)

You are a senior engineer producing ONE isolated PR. Today is 2026-07-02.
REPO: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer (a shared checkout ? follow the safety rules exactly).
AUDIT FINDINGS YOU ARE FIXING: R1-13, R1-14, R1-01(doc half), critic DATA_IN_GCS + dropped-pool items ? Grep C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md for these IDs and read the full evidence + verifier notes before coding.
TASK BRIEF:
In C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, ONE docs-only reconcile PR: (1) docs/DOC_STATUS.md:47 cleanup-audit row ? current reality (Phase 2 nearly done; remaining P10 + P16+; add a DECISION_SNAPSHOT_REGRESSION row); (2) docs/README.md:41-42 stale blurbs; (3) docs/NEXT_STEPS.md 'Where we are' rewrite to post-06-21 state ? PRESERVE the owner NO-GO block (line ~8, #404) VERBATIM; (4) docs/BACKLOG.md:39 ByteTrack row ? done (#386) and DELETE its '<0.30.0 cap' sentence (cap already lifted, requirements.txt:18); flip only what the verifier confirmed (lines 22/24 rows are PARTIAL ? annotate rather than flip, see report R1-13 verify notes); (5) docs/CODE_CLEANUP_AUDIT_2026-06-24.md: mark D13 ? (#386), refresh the P10/D1 row status (PR #390 stale/conflicting ? to be re-cut); (6) the stale ByteTrack docstring in backend/pipeline/detectors/person_detector.py (doc-comment only ? no code change); (7) docs/DATA_IN_GCS.md §5 gap-table rows contradicting §7 (trajectories/labels migrated 2026-06-22) ? reconcile. VERIFY each edit against code/git before writing it. No behavior changes.

BRANCHING SAFETY (mandatory ? shared checkout, sibling sessions exist):
1. In the MAIN repo checkout run: git fetch origin
2. Create an ISOLATED WORKTREE: git worktree add "C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-<shortname>" -b <branch-name> origin/master   (branch explicitly FROM origin/master ? never from the current local branch; use origin/main for repos whose default is main)
3. Do ALL work inside the worktree. Before committing: git branch --show-current + git log --oneline -1 must show YOUR branch on the origin-master base.
4. Stage EXPLICIT paths only (git add <file> <file>) ? NEVER git add -A / . / -u
5. Before opening the PR: git log --oneline origin/master..HEAD must list ONLY your commits.
COMMIT/PR CONVENTIONS:
- Commit message: conventional style matching the repo's history; end the message with a blank line then: Co-Authored-By: Claude Fable 5 <noreply@anthropic.com>
- PR body: what/why, a "## Verification" section stating exactly what you ran locally and what CI covers, and end with: ? Generated with [Claude Code](https://claude.com/claude-code)
- Do NOT merge the PR. Do NOT remove the worktree (a fixer may need it). Return the PR number, branch, worktree path.
QUALITY BAR:
- Read the repo's CLAUDE.md first; for the engine also docs/CODE_MAP.md §0 (file placement) before adding any new file.
- Add/extend tests for every behavior change. Run the targeted tests you touched (python -m pytest tests/<relevant> -q) + ruff check on changed files; if mypy config exists, run mypy on

touched modules. Full suite is CI's job ? do not run it.
- Default-safe: new knobs default OFF / behavior-preserving unless the task explicitly says otherwise.
- The full audit evidence is in C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/AUDIT_REPORT_khelsutra-guru_2026-07-02.md ? Grep it for your finding IDs and READ the verifier notes; they contain corrections (exact file:line, scoping traps) you MUST honor.

If the brief's premise turns out false against current code (something already fixed, file moved), adapt honestly and say so in notes ? never force a pointless diff. If you genuinely cannot complete (env broken, auth missing), open NO PR and explain in notes with pr_number=0.

2. user (2026-07-02T04:45:47.546Z)

master...origin/master

3. user (2026-07-02T04:45:47.640Z)

CLAUDE.md

4. user (2026-07-02T04:45:52.401Z)

```
# CLAUDE.md ? project memory / agent handoff

Local, AI-powered **badminton (? racket-sports) highlight indexer + rally detector**:
FastAPI + SQLite + ffmpeg + GPU CV/AI, with paid **Gemini** as the cloud AI. Sibling
repos: `sports-data-collector`, a VLC-based `rally-annotator`.

## Read these first
- **`docs/README.md`** ? doc index. **`docs/CODE_MAP.md`** ? code navigation + data
  contracts (the cold-start map; read before touching code).
- **Authoritative owned-model roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ the
  live `docs/NEXT_STEPS.md`). These SUPERSEDE the 2026-06-07 strategy docs below where they
  disagree (licensing, Gemma-4 stance, base model, conversational-QA).
- **Strategy (post-scaffolding):** `docs/COMMERCIALIZATION.md`, `docs/PMF_MARKET_RESEARCH.md`,
  `docs/PLATFORM_ARCHITECTURE.md`, `docs/DEFERRED_HARDENING_PLAN.md`. Also `docs/ROADMAP.md`
  (offline-segmenter narrative) and `docs/BACKLOG.md`.
- **Rally-detection quality track (design merged 2026-06-13, owner-gated, implementation in progress):**
  `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ? `docs/MULTI_SIGNAL_FUSION_PLAN.md`
  (fuse shuttle + person + audio cues; the held-shuttle bug is the already-built `rally_gate.py`
  "Gate A" **not wired into production** ? cheapest win) ? `docs/EXPERIMENT_HARNESS.md` (A/B + shadow
  eval so cues land flag-gated/default-OFF with zero regression) ? `docs/ROORA_LABELING_GUIDELINES.md`
  (v8 vendor labeling spec). **`docs/NEXT_STEPS.md` has the prioritized pick-up for this track.**
- **History:** `docs/archives/` ? ADRs (`decisions/DECISIONS.md`), research, and
  **`checkpoints/`** (latest: `2026-06-strategy-foundation/`). **Archive policy:** only move
  **terminal-state** (DONE/REJECTED) work; live docs stay at `docs/` top level ? and **before
  archiving, HONESTLY update the doc first** (verify claims vs code ? flip status + a completion
  note ? update inbound refs ? keep the lineage ? then `git mv`), per the checklist + living
  lifecycle register in **`docs/DOC_STATUS.md`**.
```

Current state (June 2026)
- **Scaffolding complete**; strategy foundation + owned-model roadmap merged. **No platform/owned-model implementation has started.** A 2026-06-10 audit-driven hardening sweep landed (licensing gate, detector keying, re-ingest safety, config, eval gate, API validation, reliability, CI + test coverage) ? see git history.
- **Two tracks, both owner-gated:** the **committed next PLATFORM slice = async job model** (`DEFERRED_HARDENING_PLAN.md` #16, single-tenant); the **owned-model track** starts at a greenlit Phase-0 milestone (`OWNED_MODEL_IMPLEMENTATION_PLAN.md`). **Do not start implementation without explicit owner approval.**

Architecture in one breath
- **Pluggable registries** (ABC + `Registry` singleton): segmenters / providers / validators / sports / annotations ? see `backend/pipeline/segmenters/base.py`. *Future:* `StorageBackend` + `ComputeBackend` registries (PLATFORM §3).
- `video_id` = MD5 of the file ? `backend/utils/hashing.py::compute_video_id` (one helper).
- Typed config = `backend/config/` (Pydantic). DB schema evolves via the `PRAGMA user_version` migration ladder in `backend/infrastructure/database.py`.

Committed guardrails (non-negotiable)
- **Paid LLMs (Gemini/OpenAI/Anthropic) = inference / serving / fallback ONLY ? never a training data source** (terms/copyright). Commercial-clean licensing for EVERY dependency ? code, data, AND weights: **MIT / Apache-2.0 / BSD only** (ratified 2026-06-09, owner-agreed).
- **Owned-model base:** **Qwen3-VL (Apache-2.0)** primary, InternVL3 (MIT) fallback; **Gemma-4 is AMBER / bench-only** (Apache grant likely but Prohibited-Use posture unconfirmed ? counsel needed; do NOT build on it without sign-off). **Reject Gemma 3** (viral license). **Exclude minors** from the training pool; per-user training data needs a separate explicit opt-in.
- **v1 = rally + highlight + history + correction loop**; **gait/biometrics strictly future** (GDPR Art. 9).
- **Generic data schema**; `coach ? athlete` is an **optional overlay**, not baked in.
- **Compute = GCP ONLY (ADR-013, 2026-06-20):** GPU training + serving run on **GCP** (`deploy/gcp/`; Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute** (DO GPU not enabled on the account + not cleanly credit-funded; Paperspace is subscription-gated). Don't re-evaluate DO/Paperspace for compute without a new ADR; the `$200` DO credits go to **non-compute only**. The `deploy/digitalocean/*` scripts are kept solely as provider-agnostic Linux-GPU setup reused by GCP.

Workflow conventions
- **Branch off `master`; ONE PR per work item; NO PR stacking.** Don't open a PR unless asked.
- Tests: `python run_all_tests.py` (offline, fully mocked). Python 3.10+ (the `trackers` package requires ?3.10); `ffmpeg` required.
- Never push to the default branch directly; descriptive commit messages.
- **Substantial design/project work gets a tracked project doc** following `docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker · definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` §7, `docs/GOLDEN_REGRESSION_FIXTURES.md`.
- **Launch / "bit-flip" PRs** (a default-flip that drives a tracked project to terminal state) MUST, in the SAME PR, run the `docs/DOC_STATUS.md` §2 archival due-diligence on that

project's docs (verify-vs-code ? flip status + completion note ? update inbound refs ? keep ADR lineage ? `git mv` to `docs/archives/` ? record in DOC_STATUS §3), and fix any internal-link breakage the move causes. Enforced by `.github/PULL_REQUEST_TEMPLATE.md` + the `link-check` CI job (`python -m tools.check_md_links`).
70 - ?? Hand the heaviest machine-side checks (real video, the full GPU/WSL/WASB suite) to the owner,
71 who has the golden corpus + labelled videos; verify pure-logic paths locally.
72

5. user (2026-07-02T04:45:54.820Z)

38:[Omitted long matching line]
52:### R1-01 [P1/M] tech-debt ? badminton-highlight-indexer
160:### R1-13 [P2/S] doc-rot ? badminton-highlight-indexer
169:### R1-14 [P2/S] doc-rot ? badminton-highlight-indexer
489:[Omitted long matching line]
490:[Omitted long matching line]
495:[Omitted long matching line]
496:[Omitted long matching line]
552:- [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): DATA_IN_GCS.md §5 gap table contradicts its own §7: trajectories/labels marked 'not in bucket' though migrated 2026-06-22
598:- non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): 0.5 GB duplicate video object in the corpus bucket awaiting the prescribed verify-dupe-then-delete (hygiene, `badminton-highlight-indexer/docs/DATA_IN_GCS.md`)

6. user (2026-07-02T04:46:01.169Z)

File content (29176 tokens) exceeds maximum allowed tokens (25000). Use offset and limit parameters to read specific portions of the file, or search for specific content instead of reading the whole file.

7. user (2026-07-02T04:46:09.081Z)

30
31 ### annotator:audit
32 Small, well-kept two-frontend labeling tool: a 1497-line VLC Lua extension (vlc/rally_annotator.lua, v1.7.1) and a Chrome MV3 extension (web/, v0.1.0) sharing a byte-compatible CSV format. Both CI workflows (tests, web-extension CI) are green on HEAD (last 15 runs all success; HEAD = PR #25, 2026-06-22); no open PRs; 2 open issues (#26 CSV provenance, #22 native i18n review) ? both verified still unaddressed. Web unit coverage is enforced (95/82 floors) and there is a real-Chromium per-locale E2E. README install/run instructions are accurate (VLC copy-file install; web npm install + build path is exactly what green CI runs). Debt is concentrated in three places: the content script's always-on full-DOM polling on every page, the locked LD-i8 locale-persistence decision being documented-but-unimplemented on the web side, and machine-draft translation catalogs pending native review. The sibling non-git 'Annotation Setup' dir is NOT a duplicate/predecessor of this repo ? it is the local annotation-data workspace (golden .rallies.csv labels since migrated to GCS, ~18MB trajectory CSVs, and Roora pilot-review artifacts that appear to exist nowhere else).
33
34 ### obsreport:audit
35 Tiny, high-quality private Python lib (9 src modules, 8 commits, all authored 2026-06-07; frozen since). Release hygiene is excellent: tag v0.1.3 == local HEAD == origin/main (1f72c1c), pyproject.toml version == version.py == 0.1.3, and the committed JSON Schema is byte-identical to a fresh ReportEnvelope.model_json_schema() dump (verified read-only). 74 tests pass locally in 0.43s and the code is clean (no eval, closed Literal vocabularies, never-raise IO), but the repo has ZERO CI (gh api: 0 workflows), no issues/PRs, and 3 ruff errors sit unchecked. The engine pins it correctly to tag v0.1.3 (not a floating branch) but via the stale pre-transfer org URL (avidullu, repo now lives in Khelsutra ? works only via GitHub redirect). The worst finding is consumer-side: sports-data-collector declares `obsreport>=0.1.3` as a plain PyPI requirement while the name `obsreport` is unclaimed on PyPI (404) ? a dependency-confusion window. The engine?obsreport integration tests never run in any CI (engine CI deliberately omits the private dep; reporting tests importorskip).

nongit:dirs

Two intentional but unversioned dirs. "Annotation Setup" (4.9 GB) is the golden-corpus staging area: Golden Labelled/ (6 .rallies.csv) and Trajectories/ (20 WASB CSVs, 18 MB) are VERIFIED safely migrated to gs://khelsutra-rally-corpus (15 labels + 15 trajectories uploaded 2026-06-22, listed live), so the dominant loss risk is NOT the eval corpus ? it is the Archive/Roora pilot record (3 raw vendor videos 4.86 GB + CVAT deliverables + feedback + corrected golden labels), which exists nowhere else (0 roora objects in the bucket; the engine repo archives only the guidelines). "khelsutra-screencast" (107 MB) is the documented artifacts sink of khelsutra's committed e2e screencast harness (playwright.config.mjs writes ../../khelsutra-screencast/_raw), but it also holds 8 ad-hoc Playwright QA scripts and doc-cited DoD evidence that are unversioned; only 2 of its videos are backed up to the promo bucket, and the 5 numbered checkpoint screencasts cited in STUDIO_MEDIA_LIFECYCLE.md are absent from this machine entirely. Engine docs reference the Annotation Setup dir by three different, all-nonexistent paths, and DATA_IN_GCS.md's §5 gap table contradicts its own §7 done-markers. Overall: data-loss exposure is concentrated in one 4.9 GB vendor-pilot folder plus moderate doc-rot; no secrets found in either dir.

cross:repo

Workspace of 5 git repos (all just pulled to origin master/main, zero extra worktrees) plus 2 non-git data dirs ("Annotation Setup" 4.9G raw video/labels ? the n=6 golden labels ARE versioned inside sports-data-collector's committed data/sports_metadata.db per commit 9e6f204, so the dir is raw-media overflow; khelsutra-screencast 107M e2e artifacts). The engine<->SPA runtime API contract is coherent in source: every SPA-called endpoint exists in backend/main.py + backend/api/uploads.py, and the new fps/resolution fields landed on both sides (engine main.py:516-517 via #413; SPA types.ts:44-45 + lib/video.ts:35-36 via #140). The real cross-repo drift is the COMMITTED vendored SPA bundle (backend/ui @ khelsutra ed33a515, 2026-06-22) now 10 web/ commits behind, with a skew guard that by design only catches engine-side API_VERSION bumps. CI maturity is very uneven (engine: 4 guard lanes; khelsutra: excellent incl. coverage floor + layout/a11y; sdc: pytest-only; rally-annotator: ok; obsreport: none), and docs in both engine and khelsutra still describe the DO droplet deploy as LIVE a week after its 2026-06-25 retirement. Trees are clean except sports-data-collector (uncommitted .gitignore WIP + a stale full nested self-clone), and 20 merged branches linger across remotes/locals.

engine:issues-triage

Engine repo is active and healthy on master (last merge #413 on /api/videos; CI trusted green; ~1500 tests, coverage floor 70 enforced in both CI lanes). 13 issues open (not 14) plus one PR. Triage found one live P0 cost bug (#340: /api/process re-runs the paid Gemini segmenter for an already-processed video_id ? verified no dedup short-circuit exists in current code), the D1 god-module-split PR #390 now CONFLICTING and 6 days stale while main.py grew to 2198 lines, and several issues with partially-stale premises (#388 half-done since PR #234, #331/#326/#328 still citing the retired DO droplet, #175 referencing a design doc that does not exist in the workspace). No fully fixed-but-open issues; the debt level is moderate and mostly well-tracked, but the issue text drift means several tickets need scope-correcting comments before anyone acts on them.

engine:perf-resource

Engine repo at origin/master e410136 (clean tree). Performance/resource-lifecycle posture is mixed: the deliberately-built safeguards are genuinely wired and healthy ? SQLite goes through a single choke point with WAL + busy_timeout=30000 + connect timeout=30s (backend/infrastructure/database.py:48-57; the only sqlite3.connect in backend/, so the engine does NOT have sports-data-collector's contention gap), keep_frames defaults False with post-success frame/stage deletion in all three detector runners (config/models.py:61,98; wasb_runner.py:210; tracknet_runner.py:225; native_wasb_runner.py:315), Gemini chunk files are removed after analysis (segmenters/gemini.py:289,375-377), the stitcher clips live in a TemporaryDirectory (compiler.py:419), and no unwaited Popen exists (no zombie risk). The debt is concentrated where blocking work meets the max_workers=1 job executor: every serving-path ffmpeg call runs without a timeout (extraction, audio, trim, per-clip encode, concat) so one hung ffmpeg wedges all job processing; ExecutionPolicy/run_bounded protects only the Gemini generate call. All four known perf/lifecycle issues (#349 single-threaded streaming decode, #331 CPU-bound phases, #330 MD5 reuse, #301 scheduled retention sweep) are verified still open and unimplemented in code, and one untracked leak was found: remote-input scratch copies (rally-input-*/rally-worker-* mkdtemp dirs) are never deleted and are invisible to tools/cleanup_caches.py.

49
50 ## Confirmed findings ? round 1 (43)
51
52 ### R1-01 [P1/M] tech-debt ? badminton-highlight-indexer
53 **PR #390 (D1/P10 main.py split) is stale and CONFLICTING while main.py grew 52% since it was cut**
54 - anchor: `backend/main.py:1` · tracked in: PR #390; docs/CODE_CLEANUP_AUDIT_2026-06-24.md §6 row P10 · finder: engine:audit-doc-reconcile
55 - evidence: gh pr view 390: state=OPEN, mergeable=CONFLICTING, mergeStateStatus=DIRTY, updatedAt=2026-06-26T14:37Z, 3 files (backend/main.py, backend/orchestration.py, mypy.ini). Since then >=8 PRs modified backend/main.py (#400, #405-#409, #412, #413 per git log --since=2026-06-25 -- backend/main.py). wc -l backend/main.py = 2198 vs the audit doc's '1448 lines' claim (docs/CODE_CLEANUP_AUDIT_2026-06-24.md:279); 18 @app. routes and 45 print() calls now live in it; backend.main still mypy-quarantined (mypy.ini:15). No backend/orchestration.py exists on master.
56 - why it matters: This is the last HIGH-severity item of the cleanup audit (P10, the only non-Phase-3 ? row in its §6 tracker) and it is compounding: every new endpoint PR lands in the god-module, widening the conflict surface and making the split more expensive each week. The 3-file extraction in #390 is now unmergeable.
57 - proposed fix: Close #390 and re-cut the orchestration.py extraction fresh from origin/master (small, ~1 session), or rebase it now. Then continue the split (CLI handlers -> backend/cli/, new /api/source|/api/assets|/api/capabilities routes -> backend/api/) before more endpoints accrete. Update the audit doc's stale '~1450 lines' sizing while at it.
58 - verifier notes: Minor sharpenings only: (a) git log --since=2026-06-25 -- backend/main.py actually shows 10 commits (adds #387 repo-wide ruff format ? itself a likely source of #390's conflicts ? and #380), so ">=8 PRs" is understated; (b) the mypy quarantine block is mypy.ini:15-16 ([mypy-backend.main] header at 15, ignore_errors at 16); (c) the audit doc's "1448 lines" statement is in the §Changelog 2026-06-25 "Accuracy fixes" bullet (~line 278), and the stale "~1450" sizing also appears at lines 69, 198, and 247 ? all four spots need the size refresh the fix proposes.
59 - verifier reasoning: Every load-bearing claim reproduced. (1) `gh pr view 390` returns state=OPEN, mergeable=CONFLICTING, mergeStateStatus=DIRTY, updatedAt=2026-06-26T14:37:18Z, exactly 3 files: backend/main.py (+19/-595), backend/orchestration.py (ADDED, +635), mypy.ini (+6). (2) Current working tree (branch master, HEAD e410136 = PR #413): `wc -l backend/main.py` = 2198; 1448?2198 is +51.8% ? the claimed 52%; the 1448 baseline is documented in docs/CODE_CLEANUP_AUDIT_2026-06-24.md changelog ("main.py size '~1300' ? '~1450' (now 1448 lines)", 2026-06-25 entry, ~line 278-279). (3) Not fixed: backend/orchestration.py does not exist in the tree, and git log --since=2026-06-25 -- backend/main.py shows the file only grew ? commits for PRs #413, #412, #409, #408, #407, #406, #405, #400 (the claimed 8; the log actually lists 10 incl. #387 ruff-format and #380). (4) grep counts on backend/main.py: 18 lines matching `^@app\.` and 45 `print(` occurrences ? exactly as claimed. (5) mypy.ini:15-16 = `[mypy-backend.main]` / `ignore_errors = True` (quarantine intact). (6) Audit doc §6 tracker line 198: P10 row "D1: Split main.py god-module (~1450 lines ? smaller modules)" is ?; the only other ? row is line 205 (P16+ Phase 3 bucket), so P10 is indeed the sole non-Phase-3 open item; D1 is rated HIGH at doc line 69. Severity P1 is honest for a tech-debt audit: the concrete failure is already materialized (the 635-line extraction in #390 is unmergeable today), and it compounds ? 8 endpoint-adding PRs landed in the god-module in the week since the PR was cut, all inside a mypy ignore_errors module, so each week of delay increases rework cost and unchecked-type surface. Not owner-gated: internal refactor, no serving go-live / corpus-promotion / spend / licensing implications ("P10" here is the audit tracker row, not Studio P10). Proposed fix (close-and-recut or rebase from origin/master, then continue the split per the doc's own plan at line 247, and fix the stale ~1450 sizing) is sane and correctly scoped; effort M is right given the fix includes continuing the split beyond the re-cut.
60
61 ### R1-02 [P1/M] critical-fix ? badminton-highlight-indexer
62 **/api/process re-runs the paid Gemini segmenter for an already-processed video (re-bill on refresh/'Try again') ? still unfixed at HEAD**
63 - anchor: `backend/main.py:1172` · tracked in: Khelsutra/badminton-highlight-indexer#340 (also #330) · finder: engine:correctness
64 - evidence: backend/main.py:1171-1172 computes video_id and `was_reingest = db.get_video(video_id) is not None`, but was_reingest is only used for the report/destroy-after-confirm; there is NO reuse/skip branch ? the segmenter always runs (main.py:1251-1255). Verified against gh: issue #340 OPEN ('re-runs the full (paid) segmenter anyway'), plus #330 (MD5 segmentation reuse) OPEN.
65 - why it matters: A page refresh or SPA 'Try again' on an already-indexed video (same

MD5) re-bills Gemini (~$0.10/8-min match) and burns a multi-minute compute run for zero new information. Live-triggered once in the alpha (2026-06-22, per #340).
66 - proposed fix: In _execute_processing, when db.get_video(video_id) exists with status='processed' and existing play_segments (and the request doesn't explicitly force reprocess), return the stored segments as the job result instead of dispatching the segmenter; add a `force` flag to ProcessRequest for deliberate re-ingest.
67 - verifier notes: Severity P1 is honest: concrete failure scenario is a user-reachable one-click ("Try again" after the SPA's 10-min poll ceiling, live-triggered 2026-06-22 per #340) that silently re-bills a metered Gemini key ? and per #340 the M8 cost ledger does not meter Gemini yet, so the re-bill is invisible. Proposed fix is sane and correctly scoped: the skip condition must require status=="processed" AND existing play_segments (a prior "failed" record ? main.py:1196-1201 ? must still re-run), and the added `force` flag preserves the existing destroy-after-confirm re-ingest path (main.py:1233-1235, 1278). Not owner-gated: the fix reduces real cloud spend rather than incurring it, and the owner already specified the desired behavior + acceptance criteria in OPEN issue #340. Effort M is right (endpoint logic + ProcessRequest field + a no-Gemini-call regression test). One duplicate-tracking note: #330 covers the same reuse mechanism as an optimization; a fix should close both.
68 - verifier reasoning: Verified against origin/master at HEAD e410136 (pulled, "Already up to date"). (1) Code behaves as claimed: backend/main.py:1171-1172 computes `video_id = compute_video_id(local_video)` then `was_reingest = db.get_video(video_id) is not None`; the ONLY other use of `was_reingest` in `_execute_processing` is main.py:1327, where it is passed to `emit_indexer_report` as a report field. There is no reuse/skip/short-circuit branch: after validation passes (main.py:1210-1214 upserts status "processed"), the flow unconditionally dispatches segmentation ? cloud via `_maybe_dispatch_remote` (main.py:1241-1248) or in-process `segmenter.process_video` (main.py:1251-1255). ProcessRequest (main.py:313-324) has no `force`/`reprocess` field (only video_path, player_pool, max_frames, segmenter, locale, compute). (2) Not fixed: `git log --oneline -15 -- backend/main.py` shows only unrelated commits (#413, #412, #409, #408, #407, #406, #405, #400, ...); no idempotency/reuse change. (3) Cost claim is real in served config: config.json:26 sets `"default_segmenter": "gemini"` (the paid path is the production default; the "motion" fallback at main.py:1181-1183 is only the code-level default). (4) Tracking verified via gh: issue #340 OPEN (title and body match the finding verbatim, including the 2026-06-22 live trigger and ~$0.10/8-min cost) and #330 OPEN (MD5 segmentation reuse).
69
70 ### R1-03 [P1/M] critical-fix ? badminton-highlight-indexer
71 **/api/source re-downloads the ENTIRE remote video from the bucket on every HTTP request (each browser Range request) into a fresh, never-deleted temp dir**
72 - anchor: `backend/main.py:646` · tracked in: ? · finder: engine:correctness
73 - evidence: backend/main.py:644-651: get_source calls storage.acquire_local_input(rec['filepath'], ...) on EVERY request; storage/__init__.py:89-90: for a non-local ref this does `tempfile.mkdtemp(prefix='rally-input-')` + full fetch ? a new dir + full download per call, never removed. Proxy warming is explicitly local-only (main.py:655-657), so a remote-source video never gets a proxy. FileResponse under fastapi>=0.111 (pyproject.toml:28) serves Range ? a browser scrubbing a <video> issues many separate requests to the same URL.
74 - why it matters: Direct cloud-egress cost leak plus a disk-fill denial-of-service of the appliance, reachable through a supported flow (remote-registered assets + P8 annotate-in-UI).
75 - proposed fix: Cache the fetched local copy keyed by video_id (e.g. under <output>/proxies or a per-video workspace source/ dir) and reuse it across requests; extend proxy warming to remote sources; delete or reuse the scratch dir instead of mkdtemp-per-request.
76 - verifier notes: Path correction: the storage citation is backend/storage/__init__.py:89-90 (not storage/__init__.py). Sharper failure scenario: store a 4 GB match via POST /api/assets with an s3://gs:// medium (P4, #409), open the P8 annotate UI, scrub ~10 times ? ~40 GB bucket egress (~$4-5 real cost) and ~40 GB of never-deleted rally-input-* temp dirs per session; each seek also stalls behind a full multi-GB download, so the feature is effectively unusable for remote sources. Additional color for the fix: the engine free-space guard (require_free_bytes, #407) is not applied to this download path, and the cached copy should live under a guarded root; the finding's proposed fix (cache keyed by video_id, reuse across requests, extend proxy warming to remote) is sane and correctly scoped.
77 - verifier reasoning: backend/main.py:646 calls storage.acquire_local_input(rec["filepath"], ...) on every GET /api/source/{video_id} when no proxy exists (proxy check main.py:641-642). backend/storage/__init__.py:89-90: non-local refs get a fresh tempfile.mkdtemp(prefix="rally-input-") + full fetch per call; docstring (lines 80-82) admits the copy "relies on OS temp reclamation" (nonexistent on Windows %TEMP%), and a

repo-wide grep for "rally-input" shows no cleanup path. Remote filepaths are a supported state: POST /api/assets registers the cloud stored_uri as filepath (main.py:838, 855; comment at main.py:1680 lists gs://, s3://, gdrive://). Proxy warming is explicitly local-only (main.py:655-657, docstring 628-629 "remote-source proxying is a follow-up"), so remote sources never escape the full-fetch path. Installed fastapi 0.136.1 / starlette 1.0.0 FileResponse honors Range, so a scrubbing <video> issues many independent requests, each a full-object download. git log -15 -- backend/main.py (through #413) shows no fix after #406 introduced this. Not owner-gated: the fix is plain engineering that reduces cloud spend.

78

79   ### R1-04 [P1/M] tech-debt ? badminton-highlight-indexer

80   **Import cycles: extracted API modules reach back into backend.main globals at call time; config layer imports pipeline layer**

81   - anchor: `backend/api/job_worker.py:75` · tracked in: docs/CODE_CLEANUP_AUDIT_2026-06-24.md P5 note + D1/P10 follow-up (line 221) · finder: engine:structure

82   - evidence: AST scan of backend/ found 5 strongly-connected components. (1) backend.main <-> backend.api.job_worker <-> backend.api.uploads: job_worker.py:75,98,214,234,251 and uploads.py:47 do call-time 'import backend.main as _main' to read main.py's module globals db_instance/global_config (main.py:279-280); main.py:292 re-exports job_worker names back onto itself. Documented as a transitional shim (CODE_CLEANUP_AUDIT P5 note, line 221) but it makes main.py load-bearing for every extracted module and keeps backend.main AND backend.worker.__main__ mypy-quarantined (mypy.ini:15-23). (2) Layering inversion: backend/config/models.py:249 imports segmenter_registry from backend.pipeline.segmenters.base inside a validator, while base.py:15 (_normalize_config) imports AppConfig back from config.models ? config validation now depends on pipeline registration side effects. (3-5) eval-package cycles (golden_fixtures<->golden_real_fixtures, ablation<->ablation_pdf, annotations<->gt_loader) are lazy and explicitly documented (golden_fixtures.py:1005-1028) ? lower priority.

83   - why it matters:

84   - proposed fix: Kill the shim as the concrete step-2 of D1: move db_instance/global_config onto app.state (create_app already exists per P5) and pass config/db explicitly to job_worker/uploads, then lift backend.main and backend.worker.__main__ out of mypy quarantine. For the layering inversion, move the segmenter-name validation out of the pydantic validator into a startup check (backend/config/startup.py) so config/ no longer imports pipeline/.

85   - verifier notes: Mechanism correction to the evidence: job_worker's five call-time imports do NOT read db_instance/global_config ? job_worker already takes config/db as explicit params (job_worker.py:89,205,230,247; mypy.ini:14 notes it was lifted from quarantine) and resolves FUNCTION seams (ProcessRequest, _execute_compile, _execute_processing, JobWaiting, _arm_wake_timer, _get_executor, _MAX_DRAIN_WAKE_SEC) through the backend.main module object per the documented test-monkeypatch contract (job_worker.py:8-16, main.py:286-291). Only uploads.py:49 reads a main global (global_config). So the app.state migration for db/config is mostly about uploads + main.py itself; the harder remaining work is redesigning the monkeypatch seam contract and its tests ? effort is top-end M. Sharper failure scenario: any alternative entrypoint (new console_script, uvicorn factory) that builds the app without running main() leaves backend.main.global_config=None and uploads silently falls back to default upload dir/size limits (uploads.py:49-52), ignoring operator security config ? same class as the already-documented python -m backend.main incident (main.py:2188-2195) that hung every queued job. Layering inversion is slightly worse than claimed: the validator's lazy import (models.py:249) transitively executes segmenters/__init__.py:1-7, importing ALL concrete segmenters, so constructing AppConfig anywhere drags the full pipeline in; and SegmenterRegistry.register (base.py:66-70) re-enters AppConfig construction during registration with errors swallowed by base.py:19's bare except.

86   - verifier reasoning: Verified in current tree (HEAD e410136, pulled clean). Cycle 1: backend/main.py:292-302 imports backend.api.job_worker at module load (re-exports); job_worker.py:75,98,214,234,251 do call-time 'import backend.main as _main' (grep-confirmed all 5); uploads.py:47-49 call-time imports and reads _main.global_config; globals defined main.py:279-280, set at main.py:101-103 and 1981-1982; mypy.ini:15-16 quarantines backend.main, :22-23 backend.worker.__main__. Cycle 2: config/models.py:244-256 validator imports segmenter_registry from pipeline/segmenters/base; base.py:14-18 imports AppConfig back in _normalize_config; importing base initializes segmenters/__init__.py:1-7 which imports every concrete segmenter (motion.py:243 etc. register as import side effects), so AppConfig construction depends on pipeline import health ? and register() (base.py:66-70) re-enters AppConfig via segmenter_cls(None, {}) with failures swallowed by base.py:19 bare except. Eval cycles lazy+documented (golden_fixtures.py:1008-1013). Not already fixed: git log for the four

files shows only feature commits (#405-#413). Severity honest: the failure class already bit ? main.py:2188-2195 documents the python -m backend.main dual-module incident (backend.main.db_instance=None, drain crashed silently, jobs hung 'queued' forever), patched only for that entry path (main.py:2196-2198); uploads.py:49-52 getattr-defaults mean any entrypoint that skips main() silently runs uploads with default dir/limits. Tracked_in accurate: audit doc line 221 is verbatim the P5 shim note. Not owner-gated: only owner question is D1 phase sequencing, none of the listed gates. Fix sane: create_app exists (main.py:2184), handlers already use app.state (mypy.ini:12-13), backend/config/startup.py exists as the startup-check home.

87

88 ### R1-05 [P1/M] tech-debt ? badminton-highlight-indexer

89 **Path-jail rejects gs://,s3://,gdrive-api:// before scheme resolution ? bucket ingest impossible on the exposed alpha box (issue #327 still open)**

90 - anchor: `backend/main.py:1376` · tracked in: https://github.com/Khelsutra/badminton-highlight-indexer/issues/327 · finder: engine:security-config

91 - evidence: main.py:1376 calls `validate_input_path(req.video_path, allowed_root=allowed_root)` BEFORE `storage.resolve_input_ref` at main.py:1379; backend/utils/paths.py:45-55 `resolve_under_root` realpaths the raw URI so `gs://bucket/x.mp4` escapes the jail -> 400. Startup check startup.py:106-119 makes the jail mandatory when exposed (EXPOSED_NO_VIDEO_ROOT), so an exposed engine can never ingest from a bucket. gh: issue #327 OPEN (verified 2026-07-02), proposing scheme-aware validation.

92 - why it matters: Blocks the alpha CUJ (large-video ingest via gs:// instead of 500 MB browser uploads) and is a stale-open thread since before 06-26; the fix must be done carefully to avoid reopening the arbitrary-local-file hole the jail exists for.

93 - proposed fix: Implement #327 option (b): jail only after resolve_input_ref, applying resolve_under_root exclusively to backend=='local' refs; keep traversal/symlink/absolute rejection for local paths and add both allow (bucket URI accepted) and reject (local traversal still 400) tests, updating tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri.

94 - verifier notes: Minor corrections: (1) startup check path is backend/config/startup.py:106-119, not startup.py (lines exact). (2) The "500 MB browser uploads" figure in why_it_matters comes from issue #327's body; the repo's actual browser-upload cap is security.max_upload_mb default 4096 MB (backend/api/uploads.py:51, backend/config/models.py:339) ? 500 MB is the Gemini provider per-clip cap (backend/providers/gemini.py:23). Immaterial to the finding (multi-hour matches exceed 4 GB anyway; docs/VIDEO_LOCALITY_MODEL.md:26 "The 10 GB problem is real"). (3) Fix scope additions: there is a second resolve call-site (CLI --video-path, see tests/test_api_endpoints.py:922) ? verify whether it needs the same scheme-aware treatment; ensure unknown schemes still ValueError?400 via resolve_input_ref as the sole scheme gate. (4) Caution, not a gate: the change relaxes a deliberately pinned security contract on the exposed box; #327 records the owner's explicit want, but the option (a) vs (b) choice and the PR should get owner sign-off before landing since it alters exposed-serving posture.

95 - verifier reasoning: Personally reproduced from current code. (1) Ordering: backend/main.py:1376 calls validate_input_path(req.video_path, allowed_root=allowed_root) BEFORE storage.resolve_input_ref at main.py:1379-1381; ValueError maps to HTTP 400 at main.py:1382-1383. (2) Mechanism: backend/utils/paths.py:58-68 routes any request with allowed_root set through resolve_under_root, which os.path.realpath()s the RAW string (paths.py:47) ? 'gs://bucket/x.mp4' realpaths to a cwd-relative path, commonpath != root (paths.py:50-54) ? ValueError ? 400. The code itself documents this at main.py:1387-1389: "a configured allowed_video_root (hosted mode) rejects a non-local URI as out-of-root". (3) Pinned by test: tests/test_api_endpoints.py:903-912 asserts gs://, s3://, gdrive:// all ? 400 when allowed_video_root is set ("Pins the security contract"). (4) Exposure mandate: backend/config/startup.py:106-119 emits LEVEL_ERROR EXPOSED_NO_VIDEO_ROOT when edge_auth_enabled and allowed_video_root unset ? so an exposed engine must have the jail, making bucket ingest impossible there. (5) Not fixed: git log -15 -- backend/main.py (head e410136, #413) shows no commit addressing this; behavior present in current tree. (6) gh issue view 327: state OPEN, created 2026-06-22, title/body match the finding exactly, including the owner's explicit want ("ingest large videos via gs:// on the hosted box") and options (a)/(b). Concrete failure: exposed alpha box (edge_auth_enabled=true, allowed_video_root=/jail per boot requirement) receives POST /api/process {"video_path":"gs://bucket/match.mp4"} ? 400 "path ... escapes the allowed root"; only ingest path left is the capped browser upload (backend/api/uploads.py:51, security.max_upload_mb) ? blocks the tracked alpha large-video CUJ. Severity P1 honest (owner-wanted CUJ blocked, open since 2026-06-22, workaround capped/painful per docs/VIDEO_LOCALITY_MODEL.md:19-26). Proposed fix (option b: jail only backend=='local' after resolve_input_ref, keep null-byte check universal and traversal/symlink jail for local, add

allow+reject tests, update the pinned test) matches the issue's own proposal and is correctly
scoped; effort M is right.
96
97   ### R1-06 [P1/S] doc-rot ? khelsutra (avidullu/khelsutra)
98   **Docs claim the retired droplet alpha is LIVE (droplet deleted 2026-06-25)**
99   - anchor: `NEXT_STEPS.md:12` · tracked in: partially in issue #97 (item 2: update
DEPLOY_ALPHA.md); the NEXT_STEPS.md droplet claim is untracked · finder: khelsutra:issues-triage
100   - evidence: NEXT_STEPS.md:12-13 ('Where we are (updated 2026-07-01)') still says the
site is 'live + browsable on the DO droplet -> http://168.144.126.176:8787';
deploy/DEPLOY_ALPHA.md:3 says 'Status: LIVE (deploy live-verified on the droplet 2026-06-22)'
and :29 'This runbook ships the split topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The
alpha stand-up is now LIVE (2026-06-22)'. Project memory (MEMORY.md:63, session-handoff.md:19):
droplet claude-do-rally-test/168.144.126.176 RETIRED+deleted 2026-06-25; alpha paused pending
Cloud Run. nslookup shows api./app.khelsutra.guru still resolve via Cloudflare to a tunnel with
no origin (pending CNAME/Access cleanup).
101   - why it matters:
102   - proposed fix: One doc PR: mark DEPLOY_ALPHA.md status RETIRED/PAUSED with a pointer to
the Cloud Run plan, fix NEXT_STEPS.md 'Where we are' (droplet gone, site is CF-Workers-only),
and correct LOCAL_APPLIANCE_ARCHITECTURE.md:176. Also note the single-origin repoint that
superseded the split topology (issue #97). The 2026-07-01 refresh (#139) touched docs but kept
the stale claim, so this will keep misleading ramp-up reads.
103   - verifier notes: Minor anchor correction: the 'Where we are (updated 2026-07-01)'
header is NEXT_STEPS.md:5; the droplet-live claim spans lines 12-13 (the finding's line 12
anchor is correct for the URL text). Severity P1 stands: NEXT_STEPS.md's 2026-07-01 stamp is
NEWER than the handoff's 2026-06-25 correction, so ramp-up readers reconciling the two will
trust the wrong doc; worse, DEPLOY_ALPHA.md §1 (line 35) directs deploying the engine + secrets
onto 168.144.126.176 ? an IP the owner no longer controls and DO can recycle. Fix is sane, doc-
only, and not owner-gated (it records already-made owner decisions: retirement 2026-06-25 +
single-origin repoint per issue #97); one caution ? the fix should point to the Cloud Run move
as 'paused pending' rather than declare a new serving decision, which would be owner-gated. Did
not independently re-verify the finder's nslookup/DNS claim; the finding stands without it.
104   - verifier reasoning: Working tree at HEAD f9eea14: NEXT_STEPS.md:5 'Where we are
(updated 2026-07-01)' + :12-13 'live + browsable on the DO droplet ->
http://168.144.126.176:8787 (systemd khelsutra-site)'; deploy/DEPLOY_ALPHA.md:3 'Status: LIVE
(deploy live-verified on the droplet 2026-06-22...)' + :29 'This runbook ships the split
topology'; docs/LOCAL_APPLIANCE_ARCHITECTURE.md:176 'The alpha stand-up is now LIVE
(2026-06-22): the engine booted on the droplet (168.144.126.176)'. Ground truth:
memory/khelsutra-site-droplet-parity.md:3,10-14 (droplet DELETED 2026-06-25, data archived,
doctl delete done, monitor removed via PR #117), session-handoff.md:19 ('the prior LIVE alpha on
the droplet is GONE ? alpha paused pending the Cloud-Run move'), MEMORY.md:63. Not fixed: git
log shows the last commit touching all three files is 9223ba9 (#139, the 2026-07-01 refresh) and
the stale text persists at HEAD. Tracking verified: gh issue 97 is OPEN, item 2 covers only
DEPLOY_ALPHA.md (single-origin update); the NEXT_STEPS.md droplet claim is untracked. Issue #97
also confirms split topology was superseded by the 2026-06-22 single-origin repoint, so
DEPLOY_ALPHA.md:29 is doubly stale.
105
106   ### R1-07 [P1/M] critical-fix ? khelsutra (avidullu/khelsutra)
107   **Issue #111 still fully valid: refresh loses the in-flight job; progress static for
minutes**
108   - anchor: `web/src/hooks/useIngestJob.ts:117` · tracked in: khelsutra#111 · finder:
khelsutra:issues-triage
109   - evidence: useIngestJob.ts:117-128 ? job_id lives only in the poll closure (no
localStorage/URL persistence); App.tsx:71-73 seeds state only from ?video= (set on done,
App.tsx:153). A refresh mid-job lands the user back on the ingest form while the server job
keeps running. Progress is the coarse free-text job.progress (useIngestJob.ts:106); no
heartbeat/animation beyond a spinner chip (IngestPanel.tsx:337-358).

8. user (2026-07-02T04:46:17.939Z)

158   - verifier reasoning: All cited code exists and behaves as claimed in current HEAD
(e410136). backend/main.py:109-116 `_cors_origins_from_env` returns `origins or ["*"]`;
main.py:123-129 adds CORSMiddleware with allow_origins from that fn, allow_methods=['*'],
allow_headers=['*'], allow_credentials=False. main.py:152-156 TrustedHostMiddleware defaults to
loopback hosts and matches on host.split(':')[0], so a direct fetch to http://127.0.0.1:<port>
sends Host:127.0.0.1 and passes ? it only stops DNS-rebinding via attacker domains, not direct-

IP cross-origin fetches. backend/api/auth.py:125-127 resolve_tenant returns None (no auth) when edge_auth off (default), and /api/process (main.py:1355-1370) has no other auth. Mechanism is correct: wildcard CORS is load-bearing to (a) read /api/videos responses cross-origin and (b) pass the JSON-preflight for POST /api/process (main.py:1817 ? paid Gemini). /api/process 404 fast-fail (main.py:1390) needs a valid local path, but that is harvestable from the readable /api/videos index, making the cost-leak chain concrete. git log -15 -- backend/main.py shows no fix; wildcard default is live. The finding also correctly flags the internal inconsistency: bind host (main.py:174) and allowed hosts (main.py:146) both default to loopback ('widen consciously'), yet CORS alone defaults open ? an honest oversight. Proposed fix is sane and correctly scoped; not owner-gated (local-appliance secure default, no cloud-strategy/serving-go-live decision).

R1-13 [P2/S] doc-rot ? badminton-highlight-indexer

BACKLOG.md lists three shipped items as open, including a red 'blocks something' ByteTrack entry

- anchor: `docs/BACKLOG.md:39` · tracked in: ? · finder: engine:audit-doc-reconcile
- evidence: BACKLOG.md:39 marks 'supervision ByteTrack deprecation' as red/open, but the migration shipped in #386: backend/pipeline/detectors/person_detector.py:124 imports trackers.ByteTrackTracker and requirements.txt:29 pins 'trackers @ git+...@2.5.0'. BACKLOG.md:22 lists 'App-factory / lifespan' as open, but create_app() exists at backend/main.py:93 (#370). BACKLOG.md:23 says 'orphaned Gemini uploads are never cleaned', but backend/providers/gemini.py:59 defines cleanup_orphaned_files() with a startup orphan sweep (#380). Doc header says 'Last updated: 2026-06-23'.
- why it matters: This stale state has already caused real confusion: the project memory feeding this very audit listed D13 (ByteTrack) as a remaining item. A red-flagged 'blocks something' entry that is actually done distorts session-start prioritization for every future session.
- proposed fix: Docs-only PR: flip the three entries to done with PR links (#386, #370, #380), matching the checked-off convention already used at BACKLOG.md:24. Can ride in the same reconcile PR as the DOC_STATUS/README fixes.
- verifier notes: Fix scoping correction: only the line-39 ByteTrack entry can be flipped fully to done (and its '<0.30.0 cap' sentence deleted ? the cap is already lifted in requirements.txt:18). Entries at lines 22 and 23 are COMPOUND and only half-shipped: line 22 bundles 'App-factory / lifespan + lazy torch-cv2 imports' ? create_app() shipped (#370) but there is no lifespan in backend/main.py (module-level app remains at main.py:90) and person_detector.py:34 still top-level-imports cv2; line 23 bundles 'WSL temp/frame/cache + Gemini remote-file GC' ? the Gemini GC shipped (#380) but no retention/cleanup exists for staged videos, PNG frame dumps, or *_wasbcache (#407 is a free-space guard, not GC). Correct fix: check off line 39 with #386; on lines 22-23 mark the shipped clause done with its PR link and keep the residual halves as the open item. Sharper failure scenario: the doc's own line 3 declares it 'the single tracker... so nothing gets lost between sessions', and the project memory feeding this audit already re-listed D13 as remaining ? a future session acting on the red ? line-39 entry would re-do or wrongly block a supervision bump that is already safe.
- verifier reasoning: docs/BACKLOG.md:39 marks 'supervision ByteTrack deprecation' ? open and claims 'requirements.txt is pinned <0.30.0 as a stopgap' ? but backend/pipeline/detectors/person_detector.py:124 imports `from trackers import ByteTrackTracker` (docstring lines 108-110 records the D13/P12 migration), requirements.txt:29 pins `trackers @ git+https://github.com/roboflow/trackers.git@2.5.0`, and requirements.txt:18 is now `supervision>=0.21.0` (the <0.30.0 cap the entry describes is already lifted). Commit 3624a8d (#386) shipped it. docs/BACKLOG.md:22 lists app-factory as open but create_app() exists at backend/main.py:93 (commit de09ec8, #370). docs/BACKLOG.md:23 says orphaned Gemini uploads 'are never cleaned' but backend/providers/gemini.py:59 defines cleanup_orphaned_files() with a startup orphan sweep (commit cd5b3d2, #380). Header docs/BACKLOG.md:8 reads 'Last updated: 2026-06-23'; git log shows the last BACKLOG-touching commit is b3c6c2b (#347), predating all three feature PRs. Repo is at origin/master (e410136). Not owner-gated: docs-only reconcile of factual shipped state.

R1-14 [P2/S] doc-rot ? badminton-highlight-indexer

DOC_STATUS register + docs/README index are ~15 PRs behind the cleanup-audit and decision-snapshot trackers

- anchor: `docs/DOC_STATUS.md:47` · tracked in: docs/CODE_CLEANUP_AUDIT_2026-06-24.md §8 DoD (unchecked 'docs/README.md updated' item) · finder: engine:audit-doc-reconcile
- evidence: DOC_STATUS.md:47 says CODE_CLEANUP_AUDIT is 'Phase 1: P0/P1/P4/P6a shipped... Quick wins (Q4/Q5/Q6/Q8) next', but the audit doc itself (CODE_CLEANUP_AUDIT_2026-06-24.md:8, :180-205) shows Phase 1 COMPLETE plus Phase-2 items

P8/P9/P11-P15/P17 all merged (#373-#386). DOC_STATUS has no row at all for
DECISION_SNAPSHOT_REGRESSION.md (created 06-25, merged as #383) despite §0 saying to update on
every lifecycle change; DOC_STATUS was last touched 2026-06-30 (#403) without fixing this.
docs/README.md:41 still calls the audit 'DRAFT, owner-gated' and README.md:42 says the snapshot
doc's 'decisions in §2 need locking' though D1-D11 were LOCKED 2026-06-25
(DECISION_SNAPSHOT_REGRESSION.md:26). NEXT_STEPS.md:6 still says 'owner M0 design sign-off
pending', contradicted by its own line 14 ('M0-M9 LARGELY SHIPPED... ADR-014 ratified').
173   - why it matters: DOC_STATUS is by convention the archival/lifecycle board and README is
the ramp-up index; both are read at every session start. When they contradict the per-doc
source-of-truth trackers, sessions either relitigate finished work or trust stale gating claims.
The cleanup audit's own DoD (line 225) is blocked on exactly this README update.
174   - proposed fix: One docs-only reconcile PR: refresh DOC_STATUS.md:47 row (Phase 2 nearly
done, remaining = P10 + P16+), add a DECISION_SNAPSHOT_REGRESSION row, fix docs/README.md:41-42
blurbs, and rewrite the NEXT_STEPS.md:6 'Where we are' paragraph to the post-06-21 state.
175   - verifier notes: Two refinements to the fix scope: (a) when rewriting NEXT_STEPS.md:6,
preserve the owner NO-GO block (lines 8) added by #404 verbatim ? it is a locked owner decision,
not rot; (b) DOC_STATUS.md:33 (COMPUTE_DECOUPLED_SERVING row, "M1?M4 SHIPPED ... M5 truly-local
? M6 Cloud Run ? M8 ledger ? M9-auth next") is stale for the same reason ? NEXT_STEPS.md:14 says
M5 #286/M6 #287/M8 #288/M9-auth #290 are merged ? so the same reconcile PR should refresh that
row too. Sharper failure scenario: DOC_STATUS §0 designates itself the board sessions read to
decide "what needs attention"; a session acting on row 47 would branch to implement Q4 (builtin-
defaults cache), which merged as #358 on 2026-06-25, producing a duplicate/conflicting PR.
176   - verifier reasoning: Every cited fact reproduces in the current working tree (master ==
origin/master, HEAD #413). (1) docs/DOC_STATUS.md:47 still reads "IN PROGRESS ? Phase 1:
P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next", while
docs/CODE_CLEANUP_AUDIT_2026-06-24.md:8 says "Status: PHASE 1 COMPLETE ? 14 items shipped" and
its §6 tracker (lines 180-205) shows P6b-P7 plus Phase-2 items P8/P9/P11/P12/P13/P14/P15/P17 all
? merged (#358-#386); only P10 and P16+ remain ? ? the row is ~17 merged PRs stale. (2)
DOC_STATUS.md has no row anywhere (§3a-§3e read in full) for DECISION_SNAPSHOT_REGRESSION.md,
which exists (created 2026-06-25, merged in #383 per git log of docs/README.md: commit 1283e18)
? violating DOC_STATUS.md:4's own cadence ("whenever a doc changes lifecycle state (created
...)"). (3) git log confirms DOC_STATUS.md was last touched by #403 (6e973af, the 2026-06-30
rally-cluster fix) without fixing either ? so NOT already fixed; the cleanup-audit row was last
updated in #357. (4) docs/README.md:41 still calls the audit "? Tracked project (DRAFT, owner-
gated)" and :42 calls the snapshot doc "DRAFT, owner-gated ? decisions in §2 need locking",
contradicted by DECISION_SNAPSHOT_REGRESSION.md:8 and :26 ("D1?D11 LOCKED 2026-06-25"). (5)
docs/NEXT_STEPS.md:6 says "owner M0 design sign-off pending", contradicted by line 14 in the
same file ("M0?M9 LARGELY SHIPPED (2026-06-21, ADR-014 ratified)"); #404 (the last NEXT_STEPS
commit) added the NO-GO block but left the stale paragraph. (6) The audit's own DoD at
CODE_CLEANUP_AUDIT_2026-06-24.md:225 has the unchecked "[ ] docs/README.md updated to reflect
the project's terminal state" item, matching tracked_in. Not owner-gated: a docs-only reconcile
records already-shipped, already-merged state ? no launch, corpus-promotion, strategy, spend, or
counsel decision is being made (the #404 NO-GO block must simply be preserved verbatim).
Concrete failure: a fresh session following the session-start convention reads DOC_STATUS:47,
plans "Q4/Q5/Q6/Q8 next" (shipped 5+ days ago in #358-#361), or reads README:42 and re-litigates
D1-D11 that are LOCKED ? exactly the relitigation failure the register exists to prevent. P2/S
are honest: it burns ramp-up time and risks duplicate work but breaks no code.
177
178   ### R1-15 [P2/M] test-gap ? badminton-highlight-indexer
179   **M4 quality-floor gate still missing ? the one unblocked item keeping the golden-
fixtures track from DONE**
180   - anchor: `docs/GOLDEN_REGRESSION_FIXTURES.md:156` · tracked in:
docs/GOLDEN_REGRESSION_FIXTURES.md §7 M4; docs/BACKLOG.md:60 · finder: engine:audit-doc-
reconcile
181   - evidence: GOLDEN_REGRESSION_FIXTURES.md:156 (M4 row) is ?; verified
tests/test_golden_quality_floor.py and eval_baselines/regression_baseline.json do not exist (ls:
No such file). §9 DoD (line 197) requires M4 before the track can flip DONE. BACKLOG.md:60 calls
it 'recommended-next; unblocked... the cheapest step to flip the track to DONE' (advice standing
since ~06-16).
182   - why it matters: Today the golden suite is characterization-only: it catches behavior
CHANGES but has no committed F1 floor or windowing fingerprint, so a deliberate re-freeze after
a SERVED retune could silently lock in worse quality. It is also the sole unblocked blocker to
archiving the whole fixtures project.
183   - proposed fix: Implement M4 per the doc's own recipe: seed per-slug floors from the F1
already frozen in each fx_*/expected.json score (floor = current - tolerance), record a

windowing fingerprint in eval_baselines/regression_baseline.json, keep CONTROL-only, add to the golden-crossos CI matrix.

184   - verifier notes: Two sharpenings. (a) Failure scenario, concrete: after a SERVED windowing retune the owner re-freezes real fixtures via `--refresh-snapshot --reason ...`; the characterization gate parse-compares against the NEW expected.json so it passes by construction, and with no committed floor (and no windowing fingerprint to raise "incommensurable") an F1 drop from 0.59 to e.g. 0.30 on fx_r0x lands green in CI ? the doc itself flags this rubber-stamp hole at GOLDEN_REGRESSION_FIXTURES.md:141 and names the floor baseline as the mitigation. Severity context keeping it P2 (not P1): the owner-box nightly tripwire (eval_baselines/nightly_baseline.json + scripts/nightly_regression.py) partially covers quality regressions, but only on the local uncommitted corpus, not on the committed fixtures / contributor CI path. (b) Fix note: `backend/eval/regression.py:34` already reserves DEFAULT_BASELINE_PATH = eval_baselines/regression_baseline.json for the DB-backed full-chain regression ? the M4 implementation should either share that schema deliberately or pick a distinct filename to avoid two tools silently contending for one file. Also, the finding's title should not be read as "last blocker to DONE": the P0 video-name-rename remainder (BACKLOG.md:61, GOLDEN_REGRESSION_FIXTURES.md:151 status ?) is a second DoD blocker, though owner-gated ? the finding's "sole UNBLOCKED blocker" phrasing is accurate.

185   - verifier reasoning: Every factual claim reproduced in the current tree (master, up to date with origin/master per `git status -sb`). (1) docs/GOLDEN_REGRESSION_FIXTURES.md:156 ? M4 row ("Quality-floor gate + first `regression_baseline.json`") status ?, PR "?", Depends on M1/M2 which are ? (lines 152-153), Counsel-gated "No" ? unblocked. (2) docs/GOLDEN_REGRESSION_FIXTURES.md:197 ? §9 DoD first checkbox: "**P0, M1?M4, P1, DOC** are ? and merged" ? M4 gates DONE. (3) Artifacts absent: `ls eval_baselines/` shows experiment_leaderboard.json, fixtures/, gemini_wrapper_2026-06-10.json, heuristic_lovo_n6.json, nightly_baseline.json, reelcount_manifest.json ? NO regression_baseline.json; `ls tests/ | grep -i golden` shows test_golden_fixtures.py, test_golden_fixtures_split.py, test_golden_real_fixtures.py ? NO test_golden_quality_floor.py; repo-wide grep for "quality_floor" hits only docs. (4) docs/BACKLOG.md:60 ? M4 entry marked "recommended-next; unblocked" and "the cheapest step to flip the track to DONE", exactly as cited. (5) Not already fixed: `git log --oneline -- docs/GOLDEN_REGRESSION_FIXTURES.md` tops out at 7a40434 (P3 fixtures, #317) ? no M4 commit; eval_baselines history (#202/#217/#239 nightly work) created `nightly_baseline.json`, which is a DIFFERENT mechanism (its own `_doc`: "Tied to the LOCAL golden corpus (not committed)", owner-box `scripts/nightly_regression.py`) and does not satisfy M4's committed video-free floor. (6) Failure scenario is real per the doc's own text: §6 line 141 admits `--refresh-snapshot` "can rubber-stamp a real regression" and names the separate floor-baseline update (= M4) as the mitigation; §4a line 82 confirms fixtures pin raw (vt,mg,mwd) so only M4's windowing fingerprint would flag a post-retune re-freeze as incommensurable. Not owner-gated: M4 row says counsel-gated "No", BACKLOG says unblocked/revisit-anytime; none of the owner-gate categories apply (the P0 rename remainder at BACKLOG.md:61 is the owner-gated sibling, and the finding correctly scopes itself to "the one UNBLOCKED item").

186

187   ### R1-16 [P2/L] tech-debt ? badminton-highlight-indexer

188   **Decision-snapshot regression track: decisions locked 06-25, zero code shipped since; tracker P0 row also stale**

189   - anchor: `docs/DECISION_SNAPSHOT_REGRESSION.md:66` · tracked in: docs/DECISION_SNAPSHOT_REGRESSION.md §7 (P1-P8) · finder: engine:audit-doc-reconcile

190   - evidence: §7 tracker (lines 63-73): P1-P8 all ?. Verified in code: grep 'build_stitch_plan' across backend/ and tests/ returns nothing (P2's pure stitch-plan seam was never factored out of backend/pipeline/stitcher/compiler.py), and tests/test_golden_fixtures.py has no stitch-plan snapshot. Minor rot: the P0 row (line 65) still shows ? '(this PR)' though the doc merged as #383 (commit 1283e18, 2026-06-25) and the header still says 'DRAFT'.

191   - why it matters: This gate exists precisely to bless genuine (non-AST-identical) refactors as quality-neutral ? i.e., exactly the class of change as the stalled #390 main.py split. Executing P1-P4 first would de-risk redoing the god-module split; leaving it a week-stale locked-decisions doc means refactors continue with no behavioral proof.

192   - proposed fix: Start §7 P1+P2 (canonical decision-snapshot schema + factor pure build_stitch_plan() out of compiler.py) as the next Claude-doable engine work; flip the P0 row to ? #383 and status to IN PROGRESS in the same PR. Ideally land P1-P4 before re-cutting the main.py split.

193   - verifier notes: One factual correction to the evidence: P8 (line 73) is ? (blocked/owner-gated), not ? ? "P1-P8 all ?" should read "P1-P7 all ?, P8 ?". Substance unaffected: zero of nine tracker rows show any progress. Sharper failure scenario: #390 gets re-cut and merged as a "quality-neutral" god-module split with no behavioral proof ? the only

offline gate today is the segmentation-only characterization snapshot
(test_replay_matches_committed_expected), so a subtle behavior change in stitch/concat-plan
construction (clip order, cut points, padding) ships undetected until a GPU nightly or a user-
visible reel regression. Effort L is honest for the proposed P1-P4 scope; the minimal
P0-row/status flip alone is S and could be split out.
194   - verifier reasoning: Personally verified in the current working tree (master, clean).
(1) docs/DECISION_SNAPSHOT_REGRESSION.md:8 still reads "Status: `DRAFT` ... LOCKED (2026-06-25)
... Last updated: 2026-06-25"; line 65 P0 row is "? | (this PR)" even though the doc merged as
PR #383 (gh: MERGED 2026-06-25T17:47:44Z; sole commit 1283e18 touching the file ? `git log
--oneline -15 -- docs/DECISION_SNAPSHOT_REGRESSION.md` returns exactly one commit, so nothing
fixed it since). (2) Zero code shipped: grep for build_stitch_plan across the repo hits only the
doc itself and docs/README.md; backend/pipeline/stitcher/compiler.py:24 contains only `class
VideoCompiler` with no pure plan-builder seam; grep -i stitch under tests/ returns 11 files NOT
including tests/test_golden_fixtures.py, confirming no stitch-plan snapshot exists; repo-wide
grep for stitch_plan|decision_snapshot matches only docs. Tracker rows at lines 66-72 (P1-P7)
are all ? with PR "?". (3) Why-it-matters context holds: gh shows PR #390 ("refactor(D1/P10):
extract orchestration.py from main.py (god-module split, step 1)") is still OPEN ? exactly the
non-AST-identical refactor class this gate (D11, line 40) is designed to bless. (4) Not owner-
gated: the only owner-gated rows are P0 (already satisfied ? line 82 "[x] D1?D11 locked by the
owner (2026-06-25)") and P8 (line 73); P1-P7 are marked "Gated? No", and the proposed fix
touches only non-gated rows and none of the listed gate categories.
195
196   ### R1-17 [P2/M] tech-debt ? badminton-highlight-indexer
197   **Fetched-input scratch copies are never reclaimed anywhere (process jobs, compile jobs,
/api/assets, cloud_run marker reads) ? 'OS temp reclamation' premise is false on Windows**
198   - anchor: `backend/storage/__init__.py:89` · tracked in: ? · finder: engine:correctness
199   - evidence: storage/__init__.py:80-90: acquire_local_input docstring says the scratch
copy 'is left under the system temp dir ... and relies on OS temp reclamation'. Call sites that
leak one full-video copy per run: _execute_processing (main.py:1168-1171), _execute_compile
(main.py:1683-1686), create_asset (main.py:791-799), get_source (main.py:646).
cloud_run.py:219-222 additionally leaks a cr-read-* mkdtemp per marker/report read. No janitor
exists (grep for rally-input-/cleanup found none); engine issue #301 covers only the upload_dir
retention sweep, not temp scratch.
200   - why it matters: Slow-burn disk exhaustion on exactly the appliance the product ships
as; also inflates the free-space guard's false rejections.
201   - proposed fix: Wrap remote-input acquisition in a per-job scope that deletes the
scratch dir in a finally (worker already owns the job lifecycle), or route scratch under
output_base(cfg)/tmp with an age-based sweep at startup; make cloud_run._read_json use
TemporaryDirectory.
202   - verifier notes: Two sharpenings. (a) get_source (main.py:646) cannot use a simple
finally-delete: the FileResponse streams from the scratch file after the handler returns, so
deletion there needs the age-based-sweep option (or a keyed scratch cache reused across views) ?
the finder's second proposed option covers it; per-job finally scopes fit
_execute_processing/_execute_compile/create_asset. (b) Add a fifth leaking call site to the fix
scope: the CLI single-shot path at main.py:2056. Sharper failure scenario: with
input_backend=gdrive-api/gcs, each processed match leaks one full multi-GB copy into %TEMP% per
process job, another per compile, one per /api/assets source_uri registration (fetched solely to
hash), and one per /api/videos/{id}/source view ? until
require_free_bytes(tempfile.gettempdir()) at main.py:789/1402 starts rejecting all further
remote ingests at stage="fetch" even though the disk is full of the tool's own dead scratch.
203   - verifier reasoning: Personally verified in the current tree. (1)
backend/storage/__init__.py:80-82 docstring says the scratch copy "is left under the system temp
dir ? and relies on OS temp reclamation"; line 89 does tempfile.mkdtemp(prefix="rally-input-")
and line 90 fetches into it ? no deletion. (2) Every claimed call site leaks:
_execute_processing main.py:1168-1171 (no rmtree anywhere in main.py ? grep confirmed rmtree
exists only in eval/golden_real_fixtures.py, pipeline/stitcher/compiler.py,
pipeline/detectors/*); _execute_compile main.py:1683-1686; create_asset main.py:791-799 fetches
a full video copy just to MD5-hash it (main.py:805) then returns at 865-875 without deleting
local_working; get_source main.py:646 fetches per-request and serves it via FileResponse (line
660) ? a per-view full-video leak for remote sources. Plus a fifth site the finder missed: CLI
single-shot main.py:2056. cloud_run.py:219-222 _read_json mkdtemps "cr-read-" per marker/report
read, no cleanup. (3) No janitor: tools/cleanup_caches.py scans only WSL ~/clips, the output/
tree, and security.upload_dir (docstring lines 3-5, classify() lines 52-97) ? never
tempfile.gettempdir(); its "tmp*" rule (line 79-80) applies only inside those trees. (4) Not
fixed: git log for storage/__init__.py tops out at b6214e5 (repo-wide ruff format); gh issue 301

confirmed to cover only the T11 upload-retention scheduled job, not temp scratch. (5) Guard interplay real: require_free_bytes(tempfile.gettempdir(), ?, stage="fetch") at main.py:789 and main.py:1402 ? leaked copies shrink exactly the headroom the fetch guard checks, so accumulation converts into false fetch rejections. Windows does not auto-reclaim %TEMP% by default, so the docstring premise is false on the shipping platform. Severity P2 honest: affects only remote-input flows (local paths are identity passthrough, __init__.py:85-88), slow-burn, no data loss. Not owner-gated ? pure disk hygiene, no strategy/spend/ToS/corpus decision.
204
205   ### R1-18 [P2/S] critical-fix ? badminton-highlight-indexer
206   **GDriveApiStorage.fetch_to_local buffers the entire file in RAM (get_media().execute()); put() deletes the old object before the new upload succeeds**
207   - anchor: `backend/storage/gdrive_api.py:144` · tracked in: ? · finder: engine:correctness

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480   ### R2-5 [P1/S] ? badminton-highlight-indexer
481   **Engine CLAUDE.md 'Current state' section is ~3 weeks false ? claims platform/owned-model work 'has not started'**
482   - verdict: CONFIRMED
483   - reasoning: Personally reproduced against the working tree and git history. (1) C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\CLAUDE.md:30-31 states "**No platform/owned-model implementation has started.**"; lines 34-37 say the "committed next PLATFORM slice = async job model" and the owned-model track "starts at a greenlit Phase-0 milestone... Do not start implementation without explicit owner approval"; lines 41-42 call `StorageBackend` + `ComputeBackend` registries "*Future:*". git blame shows lines 30-37 were last written 2026-06-10 (commit 62fb2fb) and never touched since, despite CLAUDE.md itself being edited 6+ times after (e735fd4 06-13 ... af1ccbe 06-30 ? all other sections). (2) Current-tree reality contradicts every one of those claims: backend/storage/ contains a StorageBackend ABC + local/gcs/gdrive/gdrive_api/s3/capabilities implementations; backend/compute/base.py + cloud_run.py exist (ComputeBackend); backend/api/job_worker.py + tests/test_async_jobs.py exist ? the "committed next slice" async job model already shipped (8bcf6ce #339 "make /api/compile async (202 + poll)", 2026-06-22); backend/flywheel.py + backend/worker + backend/eval/served_gate_a.py exist. Owned-model track: a555887 (2026-06-20) "M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed"; 4134721 archives DECOUPLED_COMPUTE as "DELIVERED (M0-M2 + M3.5 on GCP)"; 334ad0f (#336) M11 data-flywheel; b72e6de (#319) training worker fix. 314 commits landed since 2026-06-13, through 2026-07-01 (e410136), including a recorded launch NO-GO decision 2026-06-30 (4a7bfbe #404) ? i.e., the project reached launch-consideration, the opposite of "not started". (3) No fix landed: git log -- CLAUDE.md shows the last commit touching it (af1ccbe, 2026-06-30) added only the bit-flip-PR convention + Python 3.10 line; the Current state section is untouched since 06-10. Bonus staleness in the same file: lines 18-19 claim rally_gate.py Gate A is "not wired into production" while backend/eval/served_gate_a.py and c8a493d "served Gate-A litmus" (#401) indicate a served Gate-A path now exists.
484   - notes: Sharper failure scenario: CLAUDE.md is auto-loaded project memory for every fresh agent session in this heavily agent-driven repo. A fresh session reads "No platform/owned-model implementation has started" + "Do not start implementation without explicit owner approval" and will (a) tell the owner the async job model is an un-started, gated future slice and propose designing/implementing what already exists (backend/api/job_worker.py), (b) treat StorageBackend/ComputeBackend as "Future" and re-design duplicates of backend/storage/ and backend/compute/, (c) misreport project status upward (e.g., in handoffs/audits) as pre-implementation when Gen-0 weights shipped and a launch NO-GO was formally recorded 06-30. This also directly violates the workspace's own "keep it honest ? reflects real, shipped state" convention. Correctly-scoped fix (S, one doc-only PR): rewrite CLAUDE.md "Current state" (lines 29-37) to reflect reality as of 2026-07-01 ? platform slice (async jobs, storage/compute backends, capabilities/assets endpoints) SHIPPED; owned-model M0-M2+M3.5 DELIVERED on GCP, Gen-0 weights pushed; launch NO-GO 2026-06-30 with pointer to #404/docs; fix line 41-42 ("Future" ? "implemented, see backend/storage/, backend/compute/"); fix lines 18-19 Gate-A "not wired" claim against backend/eval/served_gate_a.py; re-point "committed next slice" at whatever docs/NEXT_STEPS.md currently prioritizes. Not OWNER_GATED ? this is a factual truth-sync of docs to shipped code, the same class as the repo's prior "audit truth-sync" commits (62fb2fb, 9736a3b); no spend or strategy decision required. Round-1 claim of "~3 weeks false" is accurate: section frozen 2026-06-10, contradicting implementation landed from ~2026-06-20 onward.
485
486   ### R2-6 [P1/M] ? sports-data-collector
487   **Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute

local_path and the 4.9GB source videos are single-copy with no documented backup**
488 - verdict: CONFIRMED
489 - reasoning: Registry rot half CONFIRMED by direct observation. sports-data-collector/data/sports_metadata.db (git-tracked; rows landed in commit 9e6f204 "#19 ingest the n=6 owned golden corpus", DB last touched 9f1e792 with no path fix since): videos table has 8 status='curated' rows; the 6 owned ones (adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles; license_type=Proprietary, is_commercial=1) all carry local_path under 'C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled*.MP4' ? that directory does not exist. The folder moved to 'C:\Users\avidu\Projects\khelsutra-guru\Annotation Setup\' whose 'Golden Labelled\' now contains ONLY the six .rallies.csv label files; zero copies of the MP4s exist anywhere in the workspace (searched). The committed export artifact data/eval_export/manifest.json repeats the same 6 dead absolute paths. Code consequence at sports-data-collector/src/cli/curate.py:679-684: export of a curated video with missing local_path emits md5=None plus a warning, and src/core/manifests.py:23-26 documents md5 as "the indexer's join key" ? so any fresh export of the entire owned golden corpus loses the deterministic MD5 join to badminton-highlight-indexer. Data-loss half CONFIRMED as single-copy but the "no documented backup" wording is wrong: (a) verified live via read-only 'gcloud storage ls' (2026-07-02): gs://khelsutra-rally-corpus/videos/ has 10 objects and NONE of the six originals, while labels/ DOES hold all six (2026-06-08_adarsh_avi_singles.rallies.csv etc.) and trajectories are in the bucket ? annotation work-product is safe, source footage is not; (b) Google Drive golden folder (G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15) holds only the 7 proxy clips, not the original six; (c) the sole remaining copy is documented at F:\[Khelsutra] GoPro Hero Black 12\KhelSutra\Training\Golden Labelled\ per badminton-highlight-indexer/scratch/copy_videos_to_gcs.ps1:31-36 and docs/DATA_IN_GCS.md:153 (§6 Phase-2) ? and the F: drive is NOT currently mounted (only C: and G: exist), so even that copy's integrity is unverifiable today, and no MD5s of the six were ever persisted anywhere (eval_export/manifest.json predates the md5 field from PR #51; no proxy-registry rows).
490 - notes: Corrections to the round-1 claim: (1) Size is ~37.9 GB, not 4.9 GB ? per-clip sizes annotated in scratch/copy_videos_to_gcs.ps1:31-36 (10.9+7.7+11+6.7+1.4+0.19 GB); the ~4.9-5.2 GB figure matches the three unrelated Roora pilot videos under 'Annotation Setup/Archive/.../Annotation Videos/'. (2) "No documented backup" is false ? badminton-highlight-indexer/docs/DATA_IN_GCS.md §5/§6/§8 explicitly documents the F: archive location and a Phase-2 migration plan (tracker at doc line 246: 14/15 videos still unmigrated); but "single-copy" stands: exactly one copy of the six source videos exists, on an intermittently-mounted external HDD, unverifiable right now, with no persisted hashes. (3) Downgrade from potential P0 to P1 because labels + trajectories for all six ARE in gs://khelsutra-rally-corpus (verified), so GPU-free eval/litmus and synth-train survive a drive failure; what would be permanently lost is the proprietary source footage (no re-extraction via --trajectory-source detector, no re-annotation, no re-rendering). Sharper failure scenarios: (a) F: HDD failure = permanent loss of ~38 GB of irreplaceable owned golden footage; (b) today, 'curate export' on the owned corpus emits md5=None for all six (curate.py:679-684), silently breaking the indexer MD5 join contract added in PR #51. Correctly-scoped fix (M, cross-repo): mount F:, verify + hash the six MP4s (persist MD5s into the DB/manifest immediately ? S and decision-free), generate 1080p proxies, upload via the existing scratch/copy_videos_to_gcs.ps1 or the rclone cloud path (Phase-2, already planned), then repoint/annotate local_path (or add a gs:// video_uri column) in sports_metadata.db and refresh data/eval_export/manifest.json. Partially OWNER_GATED sub-decision only: DATA_IN_GCS.md §7.4 leaves proxies-only vs full-4K migration scope OPEN and the upload is real (though tiny, cents/month) GCS spend; the hash-and-record step and the DB path fix need no owner decision.
491
492 ### R2-7 [P2/S] ? non-git
493 **Roora vendor pilot record (paid deliverables + corrected golden labels + 3 videos) is a single unversioned local copy at risk of loss**
494 - verdict: CONFIRMED
495 - reasoning: CONFIRMED for the paid deliverables + corrected golden labels; the "3 videos" component of the claim is REFUTED (Drive copies exist). Evidence: (1) Single-copy artifacts verified on disk: raw paid CVAT deliverables at 'C:/Users/avidu/Projects/khelsutra-guru/Annotation Setup/Archive/Roora Pilot Review - 6 June 2026/' ? badminton/{badminton, badminton (1), badminton (2)}/Annotations/*.xml (~2.3 MB per-frame boxes), pickleball/*.zip (3 zips, 260 KB), tabletennis/*.zip (3 zips, 213 KB) ? plus the human-corrected golden labels 'Annotation Setup/golden_labels_badminton.csv' (2.5 KB, mtime 2026-06-06) and the rescued multisport goldens 'Annotation Setup/Trajectories/roora/{pickleball,tabletennis}_rallies.csv'. (2) Not in git: sports-data-collector tracks only the AS-DELIVERED parse (data/roora_badminton_canonical.csv, verified via git ls-files + diff ? it still has rally 2 =

0:02.503?0:32.533, rally 7 end 1:51.612, no rallies 7.5/16.5), while golden_labels_badminton.csv holds the corrections (rally 2 start?0:29, rally 7 end?1:38, missed rallies 7.5@1:46 and 16.5@3:34 added, winner?out fixes) ? those corrections exist NOWHERE else (find over C:/Users/avidu/Projects + Downloads returned only this one file). (3) Not in GCS: recursive 'gcloud storage ls -r gs://khelsutra-rally-corpus/**' contains zero objects matching roora|pilot|pickle|tabletennis|sample_long|golden_labels; labels/ holds exactly the 15 canonical corpus clips (pilot video not among them); archive/ holds only a droplet archive. (4) Not on Drive: 'G:/My Drive/Sharing' has guidelines/briefs and 'Golden Annotations' (MahadevpuraSingles only) ? no CVAT exports. (5) Outside every backup convention: badminton-highlight-indexer/docs/DATA_IN_GCS.md §5 gap table (lines 117?131) covers only the 15-clip corpus and never mentions the pilot artifacts (and its line 121/142 path 'C:\?\Annotation Setup\Collect\Trajectories\' is stale ? the folder moved, no Collect dir exists); the F: backup (memory checkpoint 2026-06-22) covered only the 7 new-clip labels; docs/archives/roora-vendor/README.md archives only guidelines+ADR, not pilot results; F: is currently unmounted. REFUTED part: the 3 pilot videos have byte-size-identical cloud copies at 'G:/My Drive/Sharing/Annotation Videos/Pilot Videos/' (Sample Long Video - Baadminton.MP4 = 2,882,861,900 B; Video 2 - Pickleball.mp4 = 1,435,276,306 B; Video 3 - Tabletennis.mp4 = 871,944,715 B ? all match local), and the local 'Annotation Videos/desktop.ini' references GoogleDriveFS.exe proving Drive File Stream provenance. Owner's own memory audit had flagged the precursor ("Roora multisport CSVs live ONLY in %TEMP%? one cleanup from loss"); the rescue moved them to Annotation Setup but they remain a single unversioned C:-disk copy.

496 - notes: Correction to the finding: the at-risk payload is NOT the 4.9 GB folder ? the videos are safe on the owner's Google Drive. What is genuinely one C:-disk failure (or one folder-cleanup) from loss is ~3 MB of paid, partly irreplaceable work product: the raw vendor CVAT exports for all 3 sports (box geometry the tracked canonical CSVs discard), the human frame-by-frame correction record (golden_labels_badminton.csv + rally_review.csv ? hours of review labor and the calibration evidence behind the vendor feedback), the pickleball/tabletennis golden CSVs backing the recorded TT-0.909/PB-0.308 multisport result, and the pilot review MD/PDF record. Vendor re-export from CVAT job 4075945 is not under owner control. Failure scenario: C: SSD failure or a cleanup of the non-git 'Annotation Setup' folder silently destroys the only record of the paid pilot and its corrected labels; nothing in DATA_IN_GCS.md's migration plan would ever pick it up because the pilot artifacts are absent from its §5 gap table. Fix (S, not owner-gated for the metadata): one 'gcloud storage cp -r' of 'Annotation Setup/Archive/Roora Pilot Review - 6 June 2026/' (excluding Annotation Videos/, already on Drive) + 'Annotation Setup/{golden_labels_badminton.csv,rally_review.csv,Trajectories/roora/}' to gs://khelsutra-rally-corpus/archive/roora-pilot-2026-06/ (archive/ prefix precedent already exists; ~3?10 MB, negligible cost), then add a Roora-pilot row to DATA_IN_GCS.md §5 and fix its stale 'Annotation Setup\Collect\Trajectories' path (lines 121/142). Including the 3 pilot videos in the bucket is the only owner-gated piece (DATA_IN_GCS §7 open decision 4, ~5 GB ? $0.10/mo) and is optional given the Drive copies.

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498 ### R2-8 [P1/S] ? badminton-highlight-indexer

499 **Upload retention sweep tool exists but nothing schedules it ? unbounded upload growth + unenforced 7-day privacy retention on any hosted deployment (#301)**

500 - verdict: CONFIRMED

501 - reasoning: Tool exists: badminton-highlight-indexer/tools/cleanup_caches.py (T11 retention sweep, --uploads-retention-days default 7.0 at ~line 407; classify() purges uploaded videos + .partial-* in the uploads zone; landed cb7394b/#299). Nothing schedules it: workspace-wide grep for cleanup_caches hits only the tool/tests/docs and an error string (backend/pipeline/detectors/base.py:162); zero references in .github/workflows/nightly.yml, deploy/cloudrun/, deploy/digitalocean/, scripts/, and zero cron/timer/sweep references in khelsutra/deploy/droplet/ (khelsutra-engine.service has no companion timer). Backend startup only sweeps Gemini remote orphans (backend/main.py:1808 _maybe_sweep_gemini_orphans), not local uploads. Issue #301 is OPEN (viewed via gh today); no scheduling commit in git log between cb7394b and master e410136. Deployment is LIVE: khelsutra/deploy/DEPLOY_ALPHA.md line 3 "Status: LIVE (deploy live-verified on the droplet 2026-06-22)"; uploads land via backend/api/uploads.py into security.upload_dir (/srv/khelsutra/incoming/uploads per runbook §2.2).

502 - notes: Sharper failure scenario: the alpha droplet is live and accepting uploads behind Cloudflare Access; without any scheduled `python -m tools.cleanup_caches --apply`, a tester's footage uploaded day 0 is still on disk day 30+ (unenforced 7-day privacy retention, an explicit owner policy per the tool docstring, 2026-06-21), and output/uploads grows unboundedly until require_free_bytes (backend/api/uploads.py:93) starts rejecting new uploads (service degradation on the small CPU droplet). Mitigations that keep this P1 not P0: runbook notes testers had not yet exercised the upload CUJ (OTP login + Gemini key rotation pending as of

2026-06-22), and the owner can run the sweep manually. Fix scope (S): do NOT re-implement the sweep ? ship a systemd timer + oneshot unit (or cron line) alongside khelsutra/deploy/droplet/khelsutra-engine.service that runs `python -m tools.cleanup_caches --apply --uploads-retention-days 7 --uploads-dir <security.upload_dir>` daily with --json output captured to journald; default-OFF/opt-in flag so local single-user installs are unaffected; document enable+verify steps in DEPLOY_ALPHA.md; one smoke test asserting the wrapper passes the right flags (tool logic already unit-tested in tests/test_cleanup_caches.py). Issue #301's acceptance criteria already spec this exactly. Partial owner gate: shipping the timer unit + docs is not gated, but *enabling* it on the live droplet is an ops action requiring the owner's SSH access, and #301 flags one open design call (whether apply should also skip uploads referenced by non-terminal jobs rows) ? recommend defaulting to the time-window-only behavior and letting the owner ratify.
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505 ## Owner-gated (flagged, no action)
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507 - [P2] badminton-highlight-indexer: **P0-rename remainder: person/venue-attributable video names still committed in code and baselines** ? Defect verified in current tree: backend/eval/rally_seg_eval.py:52-54 commits MULTICOURT_CLIPS = frozenset({"mahadevpura_2", "GX010128", "Badminton_BXH_2", "Boxhill_Doubles"}) (venue-attributable), and eval_baselines/heuristic_lovo_n6.json per_video keys are adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles (person first names + venues). git log on rally_seg_eval.py (latest 0c9c693 #395) and on the baseline (#77) shows no rename ever landed. docs/GOLDEN_REGRESSION_FIXTURES.md:151 and docs/BACKLOG.md:61 track this exact remainder as DEFERRED per the owner's safe-scope decision (2026-06-16) with trigger "before any open-sourcing / public commit" ? a product/strategy owner gate ? and the coordinated fix requires renaming the owner-held gitignored output/human_lovo_manifest.json in lockstep or the gen0 LOVO training floor breaks. Repo is currently PRIVATE (gh repo view: Khelsutra/badminton-highlight-indexer, isPrivate=true), so no live public exposure today. Real, unfixed debt, but remediation is explicitly owner-gated on the open-sourcing decision and owner-side data coordination.
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509 - [P2] khelsutra (avidullu/khelsutra): **Issue #114 stale-open: CF Access guest-login debugging against a deployment that no longer exists** ? Debt is real but the fix is an owner tracker call on the owner-gated alpha/serving track. Verified: (1) gh issue view 114 ? OPEN, createdAt == updatedAt == 2026-06-23T10:25:52Z, 0 comments; body is CF-dashboard-only triage of the Access allowlist for the droplet alpha and explicitly says "Deferred by owner 2026-06-23". (2) Cited anchor exists: deploy/DEPLOY_ALPHA.md:106 is the Allow-policy/Emails-allowlist step (line 105 is the app-domain step ? off by one). (3) Deployment gone: memory session-handoff.md:19 ? droplet claude-do-rally-test/168.144.126.176 deleted 2026-06-25, "the prior 'LIVE alpha on the droplet' ... is GONE ? alpha is paused"; api.khelsutra.guru CNAME/CF-Access flagged for cleanup. (4) Not already fixed: git log -- deploy/DEPLOY_ALPHA.md shows nothing relevant after 2026-06-23 (latest 6e4dbaf PR #113 is the unrelated GPU-VM doc); no in-repo reference to #114. (5) Nothing in-repo can fix or verify it ? the Access policy lives in the Cloudflare dashboard and the origin behind app./api. no longer exists. Fix requires owner disposition (close vs retitle), squarely under the alpha/serving go-live gate.
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512 ## Critic additions (round 1 completeness pass)
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514 - [P1/M] critical-fix: **Golden-corpus registry rot: all 6 owned 'curated' videos point at a deleted absolute path, and the 4.9GB source videos are single-copy with no documented backup** (`sports-data-collector/data/sports_metadata.db:0`)
515 - evidence: Query of the git-tracked registry: videos rows adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles (all is_commercial=1, status=curated) have local_path='C:\Users\avidu\Projects\Annotation Setup\Collect\Golden Labelled\...' and video_url=NULL. `ls /c/Users/avidu/Projects/Annotation Setup` -> 'No such file or directory'; the data now lives at 'khelsutra-guru/Annotation Setup/Golden Labelled' (folder moved, registry never updated). src/cli/curate.py:224 registers owned videos with `video_url=None, local_path=video_path` ? raw operator-supplied absolute path, no normalization or later integrity check. ShuttleSet rows are also CWD-relative with mixed separators ('./data/videos\shuttleset_1.mp4'). 'Annotation Setup/Archive' is 4.9G of source match videos; grep for gs://|bucket|backup across sports-data-collector docs/README returns nothing, while docs/MANUAL_INGESTION_POSITIONING.md:13 declares the collector the 'single canonical system-of-record' for golden rally labels.

516   - fix: Add a `curate verify` (or extend cleanup.py) that resolves every registered local_path and flags dangling rows; re-point the 6 golden rows to the current 'khelsutra-guru/Annotation Setup/Golden Labelled' location; store paths relative to a configurable corpus root instead of absolute machine paths; and back the 4.9GB Archive up to the existing GCS bucket, recording the gs:// URI in video_url so the registry has a durable pointer.
517
518   - [P2/S] ci-infra: **sports-obsreport has no CI at all ? combined with the engine's skipped reporting tests, zero reporting-stack code is gated anywhere** (`sports-obsreport/pyproject.toml:7`)
519   - evidence: `ls sports-obsreport/.github/workflows` -> no such directory (every other repo in the umbrella has workflows). The repo has 10 test suites (tests/test_customer_message.py ... test_spec.py) and a dev extra with pytest+pytest-cov, but nothing runs them on push/PR to github.com/Khelsutra/sports-obsreport (HEAD = v0.1.3, main). The engine consumes it as a tagged dependency (badminton-highlight-indexer/pyproject.toml:68) and the engine's own 17 obsreport-gated tests never run in CI (already-confirmed finding) ? so no test anywhere gates the reporting stack.
520   - fix: Add a minimal .github/workflows/ci.yml to sports-obsreport: checkout, pip install -e .[dev], pytest --cov on push/PR. This closes the library half of the reporting-stack gap independently of the engine's obsreport-lane work.
521
522   - [P3/S] hygiene: **Engine .git bloated to 281MB by ~229MB of dangling raw match-video blobs; 6.5MB yolov8n.pt is permanently in pushed history and .gitignore has no guard against re-staging raw videos** (`badminton-highlight-indexer/.gitignore:49`)
523   - evidence: `git count-objects -vH`: 4716 loose objects, 266.33 MiB loose vs only 2.61 MiB in packs; du .git = 281M. Four loose blobs are raw MP4s (ftypisom headers): 22.4MB, 31.7MB, 84.5MB, 90.2MB ? all unreachable even via `git rev-list --all --reflog` (dangling from an aborted `git add`), never gc-pruned. Separately, yolov8n.pt (6,549,796 bytes) is reachable history on origin/master (added 5257fd5, removed d19557e 'replace AGPL ultralytics YOLO'), so every clone carries the AGPL-era weights forever. .gitignore covers *.pt (line 5) but only two narrow mp4 patterns (lines 49-50: backend/static/highlights_*.mp4, *_compiled.mp4) ? nothing stops the raw-match-video staging accident from recurring.
524   - fix: Run `git gc --prune=now` in the local checkout to reclaim ~260MB, and add a broad guard (e.g. `*.mp4` with explicit `!` exceptions for any tracked fixtures, or ignore the video scratch dirs) to .gitignore. Optionally note yolov8n.pt in docs as accepted history weight; rewriting pushed history is not worth it at 6.5MB.
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526   - [P3/S] tech-debt: **obsreport dependency URL still points at the pre-transfer avidullu org ? installs survive only via GitHub's transfer redirect** (`badminton-highlight-indexer/pyproject.toml:68`)
527   - evidence: pyproject.toml:68: "obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3" (echoed in requirements.txt:35). The repo actually lives at Khelsutra/sports-obsreport ? `gh repo view avidullu/sports-obsreport` resolves to nameWithOwner Khelsutra/sports-obsreport, i.e. a transfer redirect. The redirect breaks the moment any repo is re-created under the old name, and like the already-flagged trackers pin this is a mutable tag, not a commit SHA. The local clone's own remote already uses the Khelsutra URL, so the pyproject reference is the only stale spot.
528   - fix: Update the dependency URL in pyproject.toml (and the requirements.txt comment) to github.com/Khelsutra/sports-obsreport.git and pin the v0.1.3 commit SHA instead of the tag, matching whatever convention the trackers-pin fix adopts.
529
530   - [P3/S] hygiene: **1.7GB of stale yt-dlp partial-download debris in data/videos and placeholder junk rows in the committed corpus registry** (`sports-data-collector/src/crawlers/badminton_crawler.py:231`)
531   - evidence: data/videos contains _diag.mp4.part (354MB), _t.mp4.part (516MB), _v.mp4.part (849MB) plus Frag*.part and .ytdl markers, all dated 2026-05-26 ? 1.7G total of dead partial downloads (du -ch *.part *.ytdl). The crawler's failure cleanup (badminton_crawler.py:231-246) removes only `local_path`, never yt-dlp's .part/.part-Frag*/.ytdl side files. The tracked sports_metadata.db also carries two placeholder rows with video_url='https://youtube.com/watch?v=example' (staging_match_non_comm, shuttleset_12) ? test fixtures committed into the canonical registry.
532   - fix: Delete the stale *.part/*.ytdl files; extend the crawler's failure cleanup (or cleanup.py) to glob and remove yt-dlp fragment side files for the failed target; purge or clearly mark the two 'watch?v=example' placeholder rows out of the committed registry.
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535 ## Unverified backlog (dropped at the round-1 cap; lower-ranked or duplicate)
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537 - [P2] sports-data-collector: Stale doc pointers after archive moves (cross-repo DECISIONS.md path + 6 in-code CODE_HEALTH_PLAN references)
538 - [P2] sports-data-collector: curate.py is a 1531-line monolith (CLI + interactive review + import/export/proxy/licensing)
539 - [P2] sports-data-collector: Golden-label store is a binary SQLite file tracked in git ? unmergeable across the owner's concurrent sessions
540 - [P1] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Content script runs a 3 Hz full-DOM shadow-piercing scan on every page the user visits, even with the panel hidden
541 - [P1] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Locked decision LD-i8 (web language persisted + seeded from browser language) is documented as shipped but not implemented ? panel always boots in English and forgets the choice on reload
542 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Issue #26 open and unaddressed: CSV carries zero provenance ? labels are unattributable to their source video except by filename
543 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): E2E never reloads a page: the two headline persistence behaviors (resume numbering from storage, locale persistence) have no end-to-end proof, and entrypoints/ are outside unit coverage
544 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Issue #22 still fully pending: all six translation catalogs remain machine-draft ? the declared quality gate before any public launch
545 - [P2] rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): 'Annotation Setup' is the un-versioned annotation-data workspace (not a code duplicate) and its Roora pilot-review artifacts appear to exist nowhere else
546 - [P1] sports-obsreport: Collector pins obsreport as plain PyPI requirement while the name is unclaimed on PyPI (dependency-confusion window; install broken today)
547 - [P2] sports-obsreport: Engine (and both repos' comments) reference obsreport via the stale pre-transfer org URL avidullu/ instead of Khelsutra/
548 - [P2] sports-obsreport: sports-obsreport has zero CI ? tags/pushes are cut with no automated test or lint gate
549 - [P2] sports-obsreport: Engine-obsreport integration is never exercised in any CI ? the ~12 reporting tests only ever run on dev machines
550 - [P1] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): Roora vendor pilot record (paid deliverables + corrected golden labels + 3 videos) is a single unversioned local copy
551 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): khelsutra-screencast: 8 ad-hoc QA scripts and doc-cited DoD evidence are unversioned; the 5 cited checkpoint screencasts are absent from the canonical dir
552 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): DATA_IN_GCS.md §5 gap table contradicts its own §7: trajectories/labels marked 'not in bucket' though migrated 2026-06-22
553 - [P2] non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): Stale-open #175: OneDrive golden-data sharing plan never executed and largely superseded by the GCS bucket
554 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Vendored SPA bundle in engine is 10 web/ commits stale ? engine-served UI lacks the fps display its own API just shipped
555 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Engine PR #390 (D1 main.py split) is now CONFLICTING against master and going staler
556 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Engine CLAUDE.md 'Current state' is ~3 weeks false ? claims platform/owned-model work 'has not started'
557 - [P1] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: DO droplet retired 2026-06-25 but runbooks in BOTH repos still say the deploy is LIVE on it
558 - [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: sports-obsreport has zero CI ? no .github directory at all despite shipping versioned releases

- [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Lint/type gates exist only in the engine ? sports-data-collector CI is pytest-only, plus GH-Actions version drift across repos
- [P2] khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: sports-data-collector working tree is dirty: uncommitted .gitignore WIP + a full stale nested clone of itself
- [P0] badminton-highlight-indexer (Khelsutra): #340/#330: /api/process still re-bills Gemini on re-submit of an already-processed video ? no idempotency short-circuit (live P0 cost bug)
- [P1] badminton-highlight-indexer (Khelsutra): PR #390 (D1 main.py split, step 1) is CONFLICTING and stale ? master diverged 9 commits on main.py since it was cut
- [P1] badminton-highlight-indexer (Khelsutra): #332: NSFW pre-flight gate still absent ? non-owner alpha users can route arbitrary video to Gemini (ToS/abuse + paid-garbage risk)
- [P2] badminton-highlight-indexer (Khelsutra): #327: path-jail still blocks gs://-style ingest in hosted mode ? validate_input_path runs before scheme resolution
- [P2] badminton-highlight-indexer (Khelsutra): #388 premise is half-stale: the three job-worker functions already live in backend/api/job_worker.py (since PR #234, 06-18) ? remaining scope is shim removal + test patch-target migration
- [P2] badminton-highlight-indexer (Khelsutra): #384: coverage floor 70 and offline guard lanes still duplicated verbatim between ci.yml and nightly.yml
- [P2] badminton-highlight-indexer (Khelsutra): #301: upload retention sweep tool exists but nothing schedules it ? unbounded upload growth on any hosted deployment
- [P2] badminton-highlight-indexer (Khelsutra): #349 (+#331 overlap): WASB streaming decode still forced single-threaded ? L4 GPU ~80% idle; #331 additionally needs a refresh (droplet baseline retired, its lever 3 already shipped)
- [P2] badminton-highlight-indexer (Khelsutra): #328: per-video runlog/trace endpoint still missing ? obsreport exists server-side but is not user-accessible
- [P1] Khelsutra/badminton-highlight-indexer: No timeout on any serving-path ffmpeg subprocess ? one hung ffmpeg permanently wedges the single-worker job queue
- [P1] Khelsutra/badminton-highlight-indexer: MD5 segmentation reuse still unimplemented ? every re-submit of a byte-identical video re-bills Gemini and re-pays the full CPU pipeline
- [P1] Khelsutra/badminton-highlight-indexer: WASB streaming decode still single-threaded (num_workers forced to 0), no NVDEC anywhere ? L4 GPU ~80% idle during detection
- [P1] Khelsutra/badminton-highlight-indexer: Upload retention sweep still manual ? 7-day privacy retention window unenforced without an operator (#301 open)
- [P2] Khelsutra/badminton-highlight-indexer: Remote-input scratch copies (rally-input-*/rally-worker-*) are never reclaimed ? by-design temp leak outside all cleanup tooling

Unverified P3 pool

- badminton-highlight-indexer: regression.py silently passes when no baseline exists (audit B8, still open) (critical-fix, `backend/eval/regression.py`)
- badminton-highlight-indexer: Chunked upload: concurrent/proxy-retried PUT of the same part index passes the sequence check twice and interleave-appends ? corrupt assembly detected only by part-count mismatch, forcing a full multi-GB re-upload (test-gap, `backend/api/uploads.py`)
- badminton-highlight-indexer: A failed mark_job_done/mark_job_failed write leaves a job in 'running' forever (writers swallow to False, worker ignores the return) ? pollers hang until the next restart's stale sweep (test-gap, `backend/api/job_worker.py`)
- badminton-highlight-indexer: 4 stale remote branches from merged/superseded PRs never deleted (hygiene, `.git`)
- badminton-highlight-indexer: D13 is DONE but yolo_hybrid.py docstring still claims tracking uses supervision's MIT ByteTrack (licensing-accounting text now wrong) (doc-rot, `backend/pipeline/segmenters/yolo_hybrid.py`)
- badminton-highlight-indexer: ablation PDF exporter (backend/eval/ablation_pdf.py) has zero CI coverage ? its 2 tests always skip for missing reportlab/pypdf (test-gap, `tests/test_ablation.py`)
- badminton-highlight-indexer: CI test output hides skips and slow tests: -q with no -ra / --durations anywhere (hygiene, `run_all_tests.py`)
- badminton-highlight-indexer: ruff target-version pinned to py38 while the package requires Python >=3.10 and CI runs 3.13 (hygiene, `ruff.toml`)
- badminton-highlight-indexer: JWKS fetch does not enforce https on cf_team_domain ? a http:// misconfig would fetch signing keys over cleartext (security, `backend/api/auth.py`)
- badminton-highlight-indexer: GPU-box runbook instructs pasting the raw Gemini key

inline in a shell command (lands in root bash history) (hygiene, `deploy/gcp/README.md`)
589   - badminton-highlight-indexer: eval_baselines/reelcount_manifest.json hardcodes gs://khelsutra-rally-corpus URIs outside the buckets.env single-source (tech-debt, `eval_baselines/reelcount_manifest.json`)
590   - khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/: Open UX issues #99 and #103 look overtaken by the 2026-07-01 PR wave but were never reconciled (hygiene, `khelsutra/web/src/components/VideoLibrary.tsx`)
591   - khelsutra (avidullu/khelsutra) ? SPA web/ + marketing site/: web/ SPA has no linter (no ESLint config or lint script) ? react-hooks correctness classes are unguarded (hygiene, `khelsutra/web/package.json`)
592   - khelsutra (avidullu/khelsutra): Issue #102 fully open: no &debug= URL-param handling exists in the SPA (test-gap, `web/src/App.tsx`)
593   - khelsutra (avidullu/khelsutra): Issue #72 fully open: Telugu locale not started (hygiene, `web/src/i18n/index.ts`)
594   - sports-data-collector: ingest.py records a fake placeholder video URL by default (hygiene, `src/cli/ingest.py`)
595   - rally-annotator (Khelsutra/rally-annotator; local checkout C:/Users/avidu/Projects/khelsutra-guru/rally-annotator): Local git remote still points at pre-transfer avidullu/rally-annotator instead of canonical Khelsutra/rally-annotator (hygiene, `C:/Users/avidu/Projects/khelsutra-guru/rally-annotator/.git/config`)
596   - sports-obsreport: README status section is three phases and three releases stale (doc-rot, `C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport/README.md`)
597   - sports-obsreport: _plural() mishandles common nouns (e.g. item_noun="match" renders "3 matchs") in customer-facing summaries (critical-fix, `C:/Users/avidu/Projects/khelsutra-guru/sports-obsreport/src/obsreport/render/markdown.py`)
598   - non-git dirs: "Annotation Setup" + "khelsutra-screencast" (workspace root C:/Users/avidu/Projects/khelsutra-guru): 0.5 GB duplicate video object in the corpus bucket awaiting the prescribed verify-dupe-then-delete (hygiene, `badminton-highlight-indexer/docs/DATA_IN_GCS.md`)
599   - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: 13 remote + 7 local branches across the workspace are merged/closed debris (hygiene, `khelsutra`)
600   - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: SPA comments hard-code engine main.py line numbers that are already wrong and about to get worse (doc-rot, `khelsutra/web/src/api/types.ts`)
601   - khelsutra-guru workspace (cross-repo): badminton-highlight-indexer, khelsutra, sports-data-collector, rally-annotator, sports-obsreport: Agent/contributor conventions exist only in the engine: CLAUDE.md and PR templates missing or inconsistent elsewhere (hygiene, `badminton-highlight-indexer/CLAUDE.md`)
602   - badminton-highlight-indexer (Khelsutra): #175: golden-data sharing issue references a design doc and OneDrive folder that do not exist in the workspace ? stale, needs revalidation (doc-rot, `docs/GOLDEN_REGRESSION_FIXTURES.md`)
603   - badminton-highlight-indexer (Khelsutra): #172: ship-gate F1@tIoU target remains provisional ? code correctly guards, ratification is pure owner work blocked on corpus growth (tech-debt, `training/gen0/harness.py`)
604   - Khelsutra/badminton-highlight-indexer: Duplicate ffprobe duration logic: compiler._get_video_duration re-implements utils.video.get_video_duration (hygiene, `backend/pipeline/stitcher/compiler.py`)
605   - Khelsutra/badminton-highlight-indexer: Issue #331's per-phase perf baseline was measured on the retired DO droplet ? re-baseline before spending optimization effort (doc-rot, `docs/COMPUTE_DECOUPLED_SERVING/README.md`)

10. assistant (2026-07-02T04:46:56.390Z)

Audit evidence gathered. Now let me set up the isolated worktree from origin/master.

11. user (2026-07-02T04:47:00.439Z)

Preparing worktree (new branch 'docs/reconcile-trackers-2026-07-02')
branch 'docs/reconcile-trackers-2026-07-02' set up to track 'origin/master'.
HEAD is now at e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)

12. assistant (2026-07-02T04:47:06.807Z)

Worktree is on my branch at origin/master HEAD (e410136, matching the audit's verified HEAD). Now let me read all target files.

13. user (2026-07-02T04:47:09.175Z)

```
# Documentation Status & Archival Register

> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-06-21.
> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).

---

## 0. How to use this
- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.

## 1. Lifecycle legend
`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

## 2. Archival due-diligence checklist  ? *the repeatable process (do NOT just `git mv`)*
Before a foundational/tracked doc moves to `docs/archives/`:
1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.

## 3. The register

### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
| Doc | Status | Archive-ready? | Next action |
|---|---|---|---|
| `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |
```

`RALLY_QUALITY_RESEARCH.md`	Foundation, **tracked & ACTIVE** (§7)	No	Drives the R-series; complete when §7.1 single quality number is attributed
`R2_EVAL_METRIC_DESIGN.md`	IMPLEMENTED (augmenting)	No (gates R4 + dual-eval; ablation pending)	Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster
`R1_MILESTONE_LAUNCH_REPORT.md`	LAUNCHED (#204) ? terminal record	Soon (with the R-cluster)	Keep with R-series for narrative; archive when the track wraps
`RALLY_DETECTION_GAP_CLOSURE.md`	DRAFT (owner-gated)	No	G1?G3 levers; part of the R/quality cluster
`RALLY_DETECTION_FIXES_PLAN.md`	**DONE** ? findings A/B/C resolved; P2b promoted `inactivity_temporal_density` default-ON (#396)	With the R-cluster (deferred per its header; gate = R2 tolF1 ablation + §7.1 number)	Archive with the R-series when the track wraps
`RALLY_DETECTION_CODE_REVIEW.md`	Terminal snapshot ? findings A (#394), B (#394 ? promoted #396), C (`sdc#50`) all resolved	With the R-cluster	Archive with the R-series
`PER_VIDEO_WORKSPACE.md`	DRAFT, IN PROGRESS (P0 = #264)	No	? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once)
`MULTI_SIGNAL_FUSION_PLAN.md`	P1 DONE; P2+ owner-gated	No	P3/P4 (learned fusion, audio) ? needs GPU golden features
`EXPERIMENT_HARNESS.md`	P1 IMPLEMENTED	No	A/B + promotion gate; supports the quality track
`PROMINENT_COURT_DETECTION.md`	DRAFT (owner-gated)	No	C-series; multi-track selection design
`EXECUTION_POLICY.md`	P1 SHIPPED; P3b next	No	Async job policy; reused by Cloud Run dispatch
`GOLDEN_REGRESSION_FIXTURES.md`	IN PROGRESS (owner-gated)	No	Golden eval fixtures
Data layer ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`)	LIVING (data, **no-ML**)	n/a	Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`.
`CODE_CLEANUP_AUDIT_2026-06-24.md`	**IN PROGRESS** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next.	No	Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify
`DESKTOP_APP_PLAN.md`	DRAFT (owner-gated, nothing built)	No	P0 compute-feasibility spike; relates to truly-local profile
`OWNED_MODEL_IMPLEMENTATION_PLAN.md`	PLAN/roadmap (owner-gated)	No (authoritative roadmap)	Gen-0?Gen-2; the owned-model track
`PERSONALIZATION_PLAN.md`	DESIGN ? owner-adopted; Phase-0	No	Per-user models (DB v5 store landed)
`UI_ALPHA_EXECUTION_PLAN.md`	ACTIVE	No	khelsutra UI alpha track
`I18N_PLAN.md`	Design (P0); **in implementation** (adapted)	No	Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated
`POST_166_...RISK_ASSESSMENT.md`	Plan for owner review	No	Milestones/risk doc

3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")

Doc	Status	Note
`PLATFORM_ARCHITECTURE.md`	Strategy/research	The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING
`COMMERCIALIZATION.md`	? LIVING	Strategy; carries WASB-C3
`VIDEO_LOCALITY_MODEL.md`	THINKING DOC	L0 fully-local = the truly-local profile
`HOSTING_PLAN.md`	PROPOSAL	**? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**
`UI_PROPOSAL.md`	PROPOSAL (partly shipped)	UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
`STORAGE_SHARING_MODEL.md`	DRAFT (co-design pending)	?
`ANALYZERS_AND_RECONCILERS.md`	DESIGN (owner-gated)	?
`ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md`	LIVING references	Keep current; not archive targets

3c. Reports & terminal records (archive candidates once their track wraps)

Doc	Status	Note
`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING.md` · `HARDENING_LOOP_HANDOFF.md` · `DEFERRED_HARDENING_PLAN.md`	COMPLETED ? **top-level pointer stubs** (archived copies exist)	**? confirm intentional stubs (per #165) vs finish-archiving (finding #4)**

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

3e. Already archived (`docs/archives/`)

`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

ADR-001 ? ADR-014 (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

4. Open due-diligence findings / actions (2026-06-21 review)

> **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).

1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
4. **Confirm the COMPLETED top-level stubs** (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.
5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).

93
94 ## 5. Maintenance
95 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
96

14. user (2026-07-02T04:47:09.929Z)

1 # Next steps (live) ? refreshed 2026-06-21
2
3 > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5 ## ? Where we are (2026-06-21)
6 **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not** cleared for an external/commercial launch; we stay in **development**. This is the expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.
9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]].
13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** **(1)** P3 **GPU validation + screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]`; ? native-WASB 0-rally tuned-config blocker), **(2)** P2b-install (uv/script + truly-local default config), **(3)** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.
14 1. ? **`COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
15 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)
4. **[Claude] `COMPUTE_DECOUPLED_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote_if_served_safe`, behind `flywheel.*`=OFF + a faked `consent_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `_KNOWN` + `_backend_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.
5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
6. ? **Ops-Agent codified (#291)** ? `deploy/gcp/create_gpu_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

P2 ? later / gated / deferred
9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
9b. ? **DONE (#396):** the R4 density fix (finding B / P2b in `RALLY_DETECTION_FIXES_PLAN.md`) is **promoted to default-ON**. Golden-set A/B at the **SERVED** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity_temporal_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
14. **Rename repo off "badminton"** (owner, BACKLOG).
15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
16. **PMF field validation** (owner-side, business track).

Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4): ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.

Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

Updated: 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged, owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**: ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/ coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into

`trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
Authoritative roadmap:
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory
`current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

? Next-session / GPU-box pickup (2026-06-15)
M0 in-container quick wins are COMPLETE: ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

Immediate next step ? the real P3 LOVO measurement (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

Also queued (owner-gated / pending input):
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

Leaving (bulky / M2 ? no start without owner go): ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`) behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve); the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed, sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector** (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini battery is done** (see table) ? the moat is quantified.

The quality table that now drives strategy (IoU 0.5 vs human golden labels)
Path	Mahadevpura (clean-ish)	Kushagra (multi-court)
Heuristic (velocity windowing)	0.657	0.157
Gen-0 owned head (LOVO, n=6)	0.744	0.561
Two-stage WASB+Gemini refine	0.83	?

| **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |

Reading: clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges, the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)
This is the track a desktop/local agent is slated to pick up. Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.

Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records). See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.

- **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
  `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·
  `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
- **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
- **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
- **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
  1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
  2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).
  3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
- **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
- ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.
- **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are

exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
    1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
    2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
    3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
    4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
    5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
    6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

Discipline reminder: owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

Prioritized next steps
1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
   lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
   corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
   `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
   label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
   The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
   reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
   (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
   paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
   2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
   LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
   max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
   (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
   badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
   by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
   "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
   (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

       `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
       physical-visit approach; reframed product-feedback-first per owner spec):**
       `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +
       `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6
       rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
       capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered
       decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
    6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
       deletion in AI Studio); **rename repo off "badminton"** (owner action,
       in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the battery completed.

Pointers / artifacts
- Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008 (`docs/archives/decisions/DECISIONS.md`) ·
  Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
  `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
- Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact bundle; weights gitignored,
  manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
- Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest `output/human_lovo_manifest.json`
  (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation Setup\Collect\Trajectories\` ·
  labels: `...\Collect\Golden Labelled\`.
- Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
  (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a cold/prepaid project burst-throttles
  with a MISLEADING "credits depleted" 429; ramp gradually.
- ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next milestone. Committed next
  PLATFORM slice = async job model (single-tenant).

15. user (2026-07-02T04:47:20.417Z)

- RALLY_QUALITY_RESEARCH.md ? **? Foundation doc** for closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps, the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an **adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist, sequencing, the single quality **number** attributed to the effort, and a **definition of done** (project completes when all tiers land + the number is recorded). The plan that the QUALITY_ITERATIONS.md log executes against; ties together EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS / OWNED_MODEL_IMPLEMENTATION_PLAN.
- RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not

research); §7 is the source of truth.
- PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.
- RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
- EVALUATION.md ? How we measure and improve quality.
- EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
- QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
- ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
- PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

- GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).

? Documentation status & governance
- DOC_STATUS.md ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.

Product / platform plans
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend` (what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,

in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.

- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).

- PACKAGING_PLAN.md ? **? Active tracked project (`IN PROGRESS`, 11 PRs merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ? `pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA** single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md` narrative.** §7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **? Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

- DESKTOP_APP_PLAN.md ? **? Superseded narrative (by `PACKAGING_PLAN.md`; D10 = self-hosted webserver, NOT a desktop app).** Original scoping doc (2026-06-18, `[design]`): package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of VIDEO_LOCALITY_MODEL.md; is the **`LocalDesktop` ComputeBackend** of PLATFORM_ARCHITECTURE.md §3.2; the smaller/export-clean owned model (OWNED_MODEL_IMPLEMENTATION_PLAN.md) is one answer to the compute crux.

Practical / ongoing

- CREATING_REELS.md ? **? End-user guide:** install the LIGHT-tier app ? `indexer --app` ? process a video ? pick rallies ? download a watermarked reel. Task-first usage walkthrough (web UI + the API chain), with a §6 pointer to the cloud compute option. LIGHT tier only (offline `motion` / optional Gemini; no on-device GPU or one-click cloud). Install mechanics ? root [`README.md`](../README.md); packaging roadmap ? [`PACKAGING_PLAN.md`](PACKAGING_PLAN.md).

- CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md ? **? Operator runbook:** `gs://` bucket videos ? highlight reels on the approved **L4 Cloud Run** (scale-to-zero, $0 idle). Copy-pasteable: build image ? create the GPU Job ? cloud-serving control plane ? process(cloud)?query?compile?download ? prove the path (`verify_compute_path.py`) ? $0 teardown. **Verified e2e 2026-06-23.** Raw deploy knobs ? [`../deploy/cloudrun/README.md`](../deploy/cloudrun/README.md).

- SETUP_NEW_MACHINE.md ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.

- DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:`/`F:`/Drive ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.

-

[`deploy/gcp/nightly_regression/README.md`](../deploy/gcp/nightly_regression/README.md) ? **? Nightly engine-regression OPERATIONS MANUAL (one-stop shop):** enable · run on demand · change any setting (email recipients/frequency/clips/config) · observe · pause/disable/remove. The full WASB-pipeline reel-count+cost+email nightly on an ephemeral GCP GPU VM (design/status: [`NIGHTLY_ENGINE_REGRESSION.md`](NIGHTLY_ENGINE_REGRESSION.md)). The lighter CPU cached-trajectory tripwire is `scripts/nightly_regression.py` (local Windows Scheduled Task, PR #202).
- TRACKNET_WSL_SETUP.md
- Evaluation harness and calibration drivers (in `backend/eval/`)

Data layer (collection · annotation · golden sets ? **no ML**)
- [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.

Conventions & templates
- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.

Historical documents
Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).

- CODE_AUDIT_AND_TEST_HARDENING.md ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in archives/.)
- CODE_CLEANUP_AUDIT_2026-06-24.md ? **? Tracked project (DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs, tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. §7 progress tracker is the source of truth; sibling of the completed hardening sweep above.
- DECISION_SNAPSHOT_REGRESSION.md ? **? Tracked project (DRAFT, owner-gated ? decisions in §2 need locking):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.

- **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ? [`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT_STEPS.md`).
- **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ? [`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
- Superseded: per-video output layout ? [`archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (? `PER_VIDEO_WORKSPACE.md`).

See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.

16. user (2026-07-02T04:47:21.206Z)

Backlog ? deferred items to revisit

Single tracker for work we consciously deferred, so nothing gets lost between sessions.

Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry traces to a real decision/PR/doc (no speculative work).

> Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all of these to **GitHub issues** instead (or in addition).
> Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.

Last updated: 2026-06-23.

Batch / declarative orchestration
- ? **One-shot "serving startup-script" for the GPU VM** ? `create_gpu_vm.sh` already injects a metadata `startup-script` (`deploy/gcp/ops-agent-startup.sh`), but it only installs the **Ops Agent**; standing up the live engine on the VM (`khelsutra/deploy/GPU_VM_SERVING.md`) is still manual. Propose a `deploy/gcp/serving-startup.sh` fed via `--metadata-from-file=startup-script=?` (or pasted into the GCP console) that, on boot: installs ffmpeg/cv2 libs ? pulls the engine source (from GCS) + builds the venv ? *(optional)* the WASB native env (`gpu_setup.sh` + weights) ? writes the serving `config.json`/`engine.env` ? installs `cloudflared` with the tunnel token ? starts `khelsutra-engine`. So a GPU serving box comes up **hands-off** (reuses the `deploy/gcp/nightly_regression/run_on_vm.sh` bootstrap pattern). **Why deferred ? the design hinge is SECRETS:** the Gemini key / CF-Access AUD / tunnel token must **not** sit in a plaintext startup-script (instance metadata is broadly readable by anyone with `instances.get`); they should come from **Secret Manager** (VM SA ? `secretAccessor`) or be injected post-boot. Needs that small secrets-from-metadata design + which config is baked vs fetched ? not just a script (owner asked 2026-06-23). **Revisit:** when the GPU-VM serving path is used often enough to automate.
- ? **One YAML manifest drives the whole bucket?reels run** ? today, converting `gs://` videos into highlight reels is a sequence of API calls (see [`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`](CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md)). Propose a declarative **batch manifest** (`pipeline.yaml`): **inputs** (bucket prefix/glob or explicit list), **compute** target (local/cloud), **detection** (segmenter, skip_ai_handoff), per-video **selection** rules (all / top-N longest / min duration / a query string), and **output** (bucket prefix or local dir + watermark locale) ? driven by one `indexer --batch pipeline.yaml` runner that loops process?query?select?compile?deliver, **idempotent/resumable** (skips already-done `video_id`s), reusing the existing API + the compute-picker + the compute-path provenance. **Why deferred:** needs a small design ? the manifest schema, the selection grammar, idempotency/resume, and partial-failure handling ? i.e. a tracked project doc per `PROJECT_DOC_TEMPLATE.md`, not a quick add (owner agreed 2026-06-23: ship the runbook doc now, backlog the YAML driver). **Revisit:** when hands-off batch conversion is wanted ? start with a design doc.

Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`, `NEXT_STEPS.md`)
- ? **Rename the repo off "badminton"** ? owner-emphatic "ASAP" action; the project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit:** owner action, before broad external sharing.
- ? **Quantify cost-to-Gen-2 (the largest open risk)** ? benchmark real VRAM/token/$ for the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs) **before** standing up any owned serving stack. **Revisit:** before Phase-2 commitment.
- ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized (floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4 harness.

Engineering hardening deferred from the 2026-06-10 audit sweep
- ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app` leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice (#16). **Revisit:** when the async slice starts.
- ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps, `*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy. **Revisit:** before any volume/batch processing.
- ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):** `detectors/base.py::wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and tilde expansion coexist; applied to every `~`-capable arg in both runners; `tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach

`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations (repo_dir/conda_env) if config ever becomes per-tenant.

Monetization / licensing
(source: `archives/MONETIZATION_AUDIT.md` §7, ADR-005)

- ? **Ultralytics Enterprise pricing** ? get a quote **only if** the torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently *not needed* (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering YOLOv8.
- ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the "small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus output. **Revisit:** before commercial launch.
- ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX, ActionFormer, WASB, TrackNetV3), the *code* license ? the distributed *weights/dataset* terms. Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed weights.
- ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing the product. **Revisit:** when choosing the production LLM step.

Detector swap follow-ups
(source: PR #8, ADR-005)

- ? **`supervision` ByteTrack deprecation** ? `sv.ByteTrack` is deprecated as of supervision 0.28.0 and **removed in 0.30.0**; `requirements.txt` is pinned `<0.30.0` as a stopgap. Migrate to `ByteTrackTracker` from the separate `trackers` package before lifting the cap. **Revisit:** before bumping supervision past 0.30.
- ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-tested with mocks but **not yet validated on actual footage** for detection/tracking quality. (ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
- ? **`yolo_hybrid` naming** ? the registry key, `yolo_*` config keys, and `yolo-hybrid-*` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining. Optional rename. **Revisit:** if/when a clean break is worth the migration.

Golden-set subproject
(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)

- ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief + illustrated guide in the collector repo `docs/annotation/` ? rally CSV schema; ignore dead time / include every rally (incl. faults & lets) / shot = one contact; `ending_reason` (forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for badminton & pickleball.
- ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the vendor answer key. **Unblocked.** **Revisit:** anytime.
- ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor (submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora** (Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the Roora pilot confirms quality, then build the adapter.
- ? **GS-4 oracle model** ? `AutomatedProvider` interface is stubbed (this work) with a deterministic fake. Wire a *concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be a different model lineage than production, or the regression test is blind to shared failures. **Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).
- ? **GS-5 ? regression triggers** ? wire `regress()` into CI (pre-merge gate), on-demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path. **Revisit:** after GS-4.
- ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the §4 bar; escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as calibration once the gold key exists.

52 - ? **Phase-1 video collection** ? gather the round-1 golden corpus per `docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with **? held out by match**. **Unblocked.**
53 - ? **Train/eval separation guardrails** (was `DEFERRED_HARDENING_PLAN` #13; design preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs** today, kept apart by convention only; nothing *enforces* that a video used to train the learned segmenter (`backend/eval/training.py::build_training_set` ? `distill_local`) is absent from the eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design** (from the archived DEFERRED plan §2): split unit = **match, never clip** (hold ~? of each sport as eval, per `GOLDEN_SET_PHASE1_VIDEOS.md`); add a `split: "train"|"eval"` field to the collector's `curate export` manifest (cross-repo contract, ADR-002); **[DONE ? pure-logic core, PR #171]** `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids, eval_ids)` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in `compute_video_id`) + `assert_disjoint(train_ids, eval_ids)` that raises on overlap, with `tests/test_splits.py` (overlapping ids ? `assert_disjoint` raises). **STILL DEFERRED:** wire the guard into the training entrypoint (`distill_local`/`training`) and into `run_regression.py` (baseline computed strictly on the eval split); add the collector's `curate export` manifest `split` field (above); the integration test (a deliberately-leaked manifest fails the training entrypoint). Full rationale/risks: [`archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2. **Revisit:** dedicated design session with the user (needs a collector-side manifest change).
54
55 ## Golden regression fixtures (video-free, non-attributable regression suite)
56 *(source: `docs/GOLDEN_REGRESSION_FIXTURES.md` §7 tracker / §9 definition-of-done)*
57
58 Core suite **shipped** (M1·M2·M3·MX·P1·P2·P3·DOC ? ? synthetic + 6 de-attributed real `fx_r01`?`fx_r06` fixtures replay cross-OS, leak-guarded, CONTROL-only). The two items below are the **gating remainder** of the doc's §9 definition-of-done (it flips to `DONE`/archive only when **P0 + M1?M4 + P1 + DOC** are ?); O1/O2 are explicitly optional. Tracked here so the track isn't silently treated as "done" while DoD items remain.
59
60 - ? **M4 ? quality-floor gate + first `regression_baseline.json`** *(recommended-next; unblocked).* Assert per-fixture **and** mean F1 ? a committed floor (`tests/test_golden_quality_floor.py` + `eval_baselines/regression_baseline.json`). *First move:* seed the per-slug floor from the F1 already frozen in each `fx_*/expected.json` `score` (real fixtures sit at **0.0?0.59**, so floor = `current ? tolerance`, not a global bar). **Gotchas:** record a **windowing fingerprint** in the baseline (else a SERVED retune silently passes); keep it **CONTROL-only** (learned/numpy-BLAS is non-deterministic across CPUs); add it to the `golden-crossos` CI matrix. **Revisit:** anytime ? it's the cheapest step to flip the track to DONE.
61 - ? **P0 remainder ? functional video-name rename** *(deferred per owner safe-scope, 2026-06-16).* Rename the functional names (`eval_baselines/*.json` keys, `rally_seg_eval.MULTICOURT_CLIPS`, quality-log names) ? opaque ids. *Why it's hard:* the keys feed `training/gen0/harness.py`'s LOVO floor matched to the **gitignored `output/human_lovo_manifest.json` + data files** ? rename them together or the floor breaks; also collides with the hot quality track. *First move:* put the names in a gitignored `.leakscan_private` and run `python -m tools.leak_scan --names .leakscan_private` to enumerate every occurrence, then plan the coordinated rename. **Revisit:** before any open-sourcing / public commit.
62 - ? **O1 ? protobuf + deterministic serialization** *(optional).* `rally.fixture.v1` proto3 + codegen + `protoc`/`git diff --exit-code` guard. **Revisit:** only if cross-OS **byte-stable** snapshots are wanted (JSON parse-compare suffices today).
63 - ? **O2 ? Tier-1 key-gated path** *(optional).* `scripts/fetch_golden.py` (OneDrive token) and/or SOPS+age, `RALLY_GOLDEN_DIR` wired, Tier-1 CI gated on a secret (never on fork runners). **Revisit:** only if sharing person/biometric streams (Tier-1) becomes necessary.
64
65 ## Recording best practices
66 *(source: `docs/VIDEO_RECORDING_GUIDELINES.md`)*
67
68 - ? **Validate recording hypotheses** ? empirically test which capture conditions (resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote the winners into user-facing "how to record" guidance. Only the static-camera rule is currently evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure per-condition deltas.

Code & test organization
(source: owner review comments on PR #35, 2026-06-07)

- ? **`tests/` reorganization to reduce fragmentation** ? owner wants to isolate test components on a grouping metric (by subsystem/component) so `tests/` is less fragmented, mirroring the new `backend/pipeline/audio/` + `backend/eval/audio/` layout (e.g. audio tests grouped under `tests/audio/`, eval tests under `tests/eval/`). The exact grouping **metric is the owner's design choice** (consolidate small files vs. add per-area dirs ? the "lesser fragmentation" goal slightly tensions with adding dirs), so this is left as a **dedicated refactor PR**, not guessed here. **Revisit:** when the owner specifies the grouping scheme. (The `backend/eval/audio/` half of the request ? comment 1 ? was done in PR #35.)

Commercial Readiness & Low-Regret Moves ? Approved Decisions Checkpoint (2026-06-07)

(source: Commercial Readiness Review + structure/sequencing discussion with Grok as second-in-command)

- ? **Docs-only checkpoint PR** ? This update + the refreshed `docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (since archived) formally record the approved path before any code changes. The initial review docs (PR #37) introduced the findings; this entry locks the execution decisions.
- ? **Approved directory structure** ? Adopt a layered approach under `backend/` for professionalism and future growth (no flat dumping of new modules, no explosion of 15+ single-file dirs):
  - `backend/config/` (package ? start with this)
  - `backend/api/` (FastAPI app, routes, static UI)
  - `backend/pipeline/` (validators + segmenters + stitcher + audio + detectors)
  - `backend/infrastructure/` (database + external adapters)
  - Keep `backend/eval/` prominent as a first-class concern.
  - Logical grouping for providers, sports, annotations, utils, and a small `core/` only when needed.
- ? **Sequencing (Config first, then atomic refactor)** ? Implement Deliverable 1 (Config) **first**, introducing the `backend/config/` package immediately. Perform the full directory structure refactor as a **separate high-pri atomic PR** afterwards. This minimizes conflicts, keeps PRs focused, and gives Claude a clean window with no concurrent moves.
- ? **Deliverable 1 (Config)** ? `backend/config/` package (`models.py` + `loader.py`) instead of a single `config.py`. Typed `AppConfig`, central loader with built-in defaults + file override, `extra="allow"` for migration. Update `setup.py` + `main.py`. Add tests. Land this before the layout move.
- ? **Subsequent items** ? See the refreshed `docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (archived) for the full ordered list (Error contracts, logging, packaging, detector abstraction, etc.). The structure refactor is now explicitly tracked as its own high-pri step right after Config.

Status legend reminder: ? = approved / in progress tracking, ? = do when triggered.

Observability reports ? shipped; intersections with the low-regret items (2026-06-07)
(source: per-video observability feature ? indexer PR #41/#42/#43, collector PR #11/#12; shared lib `obsreport`)

- ? **Low-regret items 1 (Config pkg, PR #39) + 2 (directory restructure, PR #40) are DONE.** Remaining = items 3?9 in `docs/archives/past_projects/low-regret-moves/LOW_REGRET_MOVES_PLAN.md` (archived).
- ? **Per-video observability reports shipped in BOTH tools** (a TECHNICAL report + a CUSTOMER-friendly summary), via the **soft-dependency** lib `obsreport` (github.com/avidullu/sports-obsreport, MIT, v0.1.3). Each run writes `<video>.report.json / .report.md / .summary.md` next to the video.
- ? **Item 8 (Input-Quality UX) is LARGELY SUBSUMED** by these reports (validation stage + footgun flags + verdict already surface quality; a clean customer summary already exists).

Do NOT build a parallel mechanism ? instead EXTEND the report: add an audio-readiness stage/flag to `backend/reporting/` (harvest) + the obsreport spec. **Revisit:** when item 8 is picked up.
- ? **Item 7 (unify hybrid segmenters) should land before the report's real AI-provider timing.** The report records AI timing as `not_captured`; once `wasb_hybrid`/`tracknet_hybrid` are unified there is ONE place to thread a recorder callback through `provider.analyze_video` ? the report. **Revisit:** with item 7 (Claude to add the timing hook there).
- ? **Item 3 (Error/Result contract) synergizes with the report.** Today the report INFERS failures (e.g. `silent_provider_noop` = 0 segments + no API key). A structured Result/error from providers/segmenters would let the report read the real reason as a first-class flag. **Revisit:** with item 3.
- ? **`obsreport` is a SOFT/commented dep** in `requirements.txt` (works locally via editable install; pipeline no-ops if absent). **Pin the `git+ssh@v0.1.3` tag** in both repos once SSH/CI access to the private lib repo exists.
- ? **`run_all_tests.py`** added at repo root ? run the full suite (clear PASS/FAIL, pytest exit code) before sending PRs.
- ? **Config-migration follow-ups (DONE ? config & identity consistency pass)** ? the two typed-config bugs from
  #39/#40 are fixed: (1) **`/api/compile` now honors `--output-dir`** and (2) **`output_dir` is a real field on
  `OutputConfig`** (default `"output"`); the CLI overrides it on the typed model in `main()`, retiring the
  write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5 derivation into
  `backend/utils/hashing.py::compute_video_id` and migrated all four `run_*.py` scripts off raw `json.load` onto
  `backend.config.load_config`. See `CODE_MAP.md` §2.0/§6 (footguns retired).

17. user (2026-07-02T04:47:31.868Z)

<!--
PROJECT DOC ? tracked code-cleanup + audit remediation plan.
Follows the PROJECT_DOC_TEMPLATE.md conventions.
-->

Code Cleanup ? 2026-06-24 Deep Audit Remediation

> **Status:** `PHASE 1 COMPLETE` ? 14 items shipped + D2 app-factory + test fixes · **Owner:** Avi · **Created:** 2026-06-24 · **Last updated:** 2026-06-25 (DoD/line-count accuracy fixes ? see Changelog)
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/past_projects/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`.
> **Relation to existing docs:** Sibling of `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (the 2026-06-10 hardening sweep).
>
> **Phase 1 summary:** All structural refactorings landed. `backend.api.job_worker` lifted from mypy quarantine. Suite: **1463/1463 green**. `ruff`: all checks passed. `mypy`: quarantined modules unchanged (ratchet held).

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0. TL;DR

A deep, whole-repo audit on 2026-06-24 surfaced **35 items** (33 from fresh review + 2 absorbed from `docs/BACKLOG.md`) across bugs, technical debt, operational risks, and quick wins. This doc organizes them into a phased cleanup project: **Phase 1 = pure refactorings** (no logic change, all tests must stay green, coverage must not drop), **Phase 2+ = logic fixes + hardening**. The first approved targets are **B2, D3, D2, B5** ? the `Success`/`Failure` contract for Gemini, typed-config migration completion, app-factory refactor, and the no-op segmenter validator fix.

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1. Problem & goal

The codebase has grown rapidly since the June 2026 scaffolding and has accumulated several patterns that make it harder to extend safely:

- **Silent failure masking** ? `analyze_video()` returns `[]` on ANY error (missing API key, network failure, model crash) indistinguishable from "video has no rallies."
- **Dual config access** ? new code uses typed `AppConfig` attribute access; old code (validators, segmenters) uses legacy `.get("indexing",{}).get(...)` dict chains. The `extra='allow'` silently keeps typo'd config keys.
- **Optional module globals** ? `db_instance` and `global_config` are `Optional[...] = None`, set only in `main()`. Importing `app` without `main()` ? NPE on every endpoint. The mypy quarantine on `backend.main` and `backend.api.job_worker` exists because of this.
- **No-op validator** ? `segmenter_must_be_registered` field validator is a pass-through; a typo'd segmenter name in `config.json` passes validation and fails at runtime.

The goal is to fix these systematically, with **zero behavior change** in Phase 1, and a **strict no-regression gate** (all tests pass, coverage never decreases).

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2. Decisions locked

#	Decision	Source / date	Implication
D1	**Phase 1 = pure refactorings only.** No logic or behavior change. All tests must pass; coverage must not decrease.	Owner, 2026-06-24	B2 (Success/Failure contract for Gemini) is deferred to Phase 2 since it changes return types callers depend on. D3 (typed-config migration), D2 (app-factory), and B5 (segmenter validator) are pure-refactoring-possible and are the first targets.
D2	**First targets: D3 ? B5 ? D2** in that order (increasing risk). D3 (typed-config migration) is the safest ? fix the validator callers one module at a time. B5 (segmenter validator) is a one-liner. D2 (app-factory) is the largest but still a refactoring. B2 (Success/Failure in Gemini) is the first Phase-2 item.	Owner, 2026-06-24	Each lands via a separate, small PR branched from `master`; no PR stacking.
D3	**Docs-only PR first.** This document lands as a single PR with no code changes. The subsequent code PRs reference this doc.	Owner, 2026-06-24	The doc is approval for the approach; individual code PRs are approval for the implementation.
D4	**Coverage ratchet:** the CI `coverage` lane must not decrease for any PR in this project.	Convention from `CODE_AUDIT_AND_TEST_HARDENING.md`	Every code PR must include updated/additive tests. Pure refactorings that remove dead code may increase coverage; a drop is a hard fail.
D5	**Follow the CODE_MAP placement rules.** No new top-level files without a placement-rule match. Tools go in `tools/`, standalone run-drivers in `scripts/`, eval library code in `backend/eval/`.	`docs/CODE_MAP.md` §0	Keeps the repo navigable as it evolves.
D6	**mypy quarantine ratchet:** when touching a quarantined module, try removing its `ignore_errors = True` entry from `mypy.ini`.	`mypy.ini`	D2 (app-factory) is the biggest opportunity ? if `backend.main` and `backend.api.job_worker` can leave quarantine, that's a major milestone.
D7	**Absorb relevant BACKLOG items.** Deferred hardening items from `docs/BACKLOG.md` that overlap with this audit are absorbed into this project rather than left in BACKLOG. Items that are feature-gaps (not code-cleanup) stay in BACKLOG.	PR #352 review, 2026-06-24	D2 and O1 were already independently found; D13 and D14 are new absorptions. The train/eval splits guardrail stays in BACKLOG (it needs a collector-side manifest change).

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3. Findings register (full audit)

3a. Bugs & Defects

ID	Severity	File(s)	Issue	Phase
B1	**HIGH**	`backend/pipeline/segmenters/motion.py`	**Fabricated data.**

Random values for server/receiver/shot_count/events/intensity/avg_speed_kmh persisted as real data. | 2 |
B2	**HIGH**	`backend/providers/gemini.py`	**Silent zero-segment returns mask API failures.** `analyze_video()` returns `([], metrics)` on ANY failure, indistinguishable from "no rallies."	2
B3	MEDIUM	`backend/pipeline/validators/pts_continuity.py`	Hardcoded 10.0s gap threshold (comment says 8.0s), not config-driven.	3
B4	MEDIUM	`backend/pipeline/stitcher/compiler.py::_parse_time`	Unparseable timestamps silently default to 0.0 ? could stitch from time zero.	3
B5	MEDIUM	`backend/config/models.py::IndexingConfig`	`segmenter_must_be_registered` validator is a **no-op pass-through**. Typo'd segmenter passes validation.	**1**
B6	LOW	`backend/pipeline/validators/resolution_fps.py`	Hard-fails if `format_check` didn't run first ? ordering is an import-time side-effect.	3
B7	LOW	`backend/eval/calibrate_wasb.py::load_golden_gts`	Silently skips bad rows (opposite of `annotations.load_annotations` which RAISES).	3
B8	LOW	`backend/eval/regression.py`	No baseline ? `regressed=False` (silent pass). Library function diverges from `run_regression.py` exit 3.	3

3b. Technical Debt / Design Smells

ID	Severity	Area	Issue	Phase
D1	HIGH	`backend/main.py`	**Massive god-module** (~1450 lines). CLI, API, worker orchestration, upload, debug-clip, trim, reingest all in one file.	2
D2	HIGH	`backend/main.py` + `backend/api/job_worker.py`	**Optional module globals** (`db_instance`, `global_config`). NPE-prone; mypy-quarantined.	**1**
D3	HIGH	`backend/config/models.py`	**Dual config access.** Typed Pydantic model + legacy `.get()` dict shim. `extra='allow'` silently keeps typo'd keys.	**1**
D4	MEDIUM	`backend/config/loader.py`	Shallow top-level merge: a `config.json` section REPLACES the defaults dict for that section ? new fields missed on upgrade.	2
D5	MEDIUM	`backend/main.py`	`print()` used in production backend code (~20 calls in `--debug-clip`). Bypasses logging.	3
D6	MEDIUM	`backend/eval/`	Duplicate "annotations" modules: `backend.eval.annotations` vs `backend.annotations`. Different schemas, different error handling.	3
D7	MEDIUM	Multiple	Hardcoded tuning constants duplicated by copy (BASELINE, TUNED, padding).	3
D8	MEDIUM	`backend/pipeline/segmenters/base.py`	Registry instantiates `cls(None, {})` to read `.name` ? `__init__` must not touch `self.db`/`self.config`. Implicit contract.	3
D9	LOW	`backend/pipeline/validators/__init__.py`	Import mutates global registry as side-effect. Removing import = de-registering validator.	3
D10	LOW	`backend/reporting/spec.yaml` + `harvest.py`	Tight coupling via magic `when.path` strings. Rename a detail ? rule silently stops firing.	3
D11	LOW	`backend/pipeline/stitcher/compiler.py`	Two-stage codec contract is implicit. Change a preset ? concat breaks.	3
D12	LOW	`tools/generate_highlights.py`	Uses `sys.path.insert(0, ...)` hack instead of proper package install.	3
D13	MEDIUM	`requirements.txt`, `pyproject.toml`	**Absorbed from BACKLOG.** `supervision` ByteTrack is deprecated as of 0.28.0 and **removed in 0.30.0**. The `<0.30.0` pin in `requirements.txt` and `pyproject.toml` is a stopgap ? migration to the standalone `trackers` package is needed before the pin can be lifted.	2
D14	LOW	`tests/` directory	**Absorbed from BACKLOG.** 90+ test files in a flat `tests/` directory. Owner wants reorganization by subsystem (e.g. `tests/eval/`, `tests/audio/`) to reduce fragmentation. The exact grouping metric is the owner's design choice.	3

3c. Operational Risk

ID	Severity	Area	Issue	Phase
O1	HIGH	`backend/providers/gemini.py`	**Orphaned remote files on process death.** No GC for uploaded videos on Gemini File API.	2

O2	MEDIUM	`backend/main.py`	Single `ThreadPoolExecutor(max_workers=1)` ? one stuck job blocks all.	2
O3	MEDIUM	`backend/infrastructure/database.py`	SQLite WAL + busy_timeout but concurrent writers (CLI + server) risk `SQLITE_BUSY`.	3
O4	LOW	`backend/pipeline/detectors/tracknet_runner.py`	`weights_path=''` ? silent zero results. No startup weights-exist validation.	3
O5	LOW	`backend/utils/single_instance.py`	Windows byte-range locks may not work on network drives (Google Drive, SMB).	3

3d. Quick Wins (Low Risk, High Value)

ID	Area	Issue	Phase
Q1	`backend/config/models.py`	Fix the no-op `segmenter_must_be_registered` validator (same as B5).	**1**
Q2	`backend/pipeline/segmenters/motion.py`	Fix return type annotation: `List[Dict]` ? `Result[List[Dict[str, Any]]]`.	1
Q3	`run_eval.py`, `run_regression.py`, `scripts/`	Deduplicate `_default_db_path()` helper into a single shared utility.	1
Q4	`backend/config/loader.py`	Cache builtin defaults dict at module level (avoid double-instantiation of `AppConfig`).	1
Q5	`requirements.txt` vs `pyproject.toml`	Add comment explaining why `torch`/`torchvision` are in requirements.txt but not `pyproject.toml`.	1
Q6	`ruff.toml`	Enable `"I"` (isort) rule ? auto-fixable, consistent import ordering.	1
Q7	`mypy.ini`	Type-clean `motion.py` (small, legacy) as a ratchet exercise.	2
Q8	`backend/main.py`	Add comment explaining lazy `import cv2` in `--debug-clip` (cv2 not in LIGHT tier).	1

3e. Absorbed from BACKLOG

These findings were already recorded in `docs/BACKLOG.md` as deferred items from the 2026-06-10 hardening sweep. They are absorbed into this project for execution:

My ID	BACKLOG item	BACKLOG status	Absorbed as
D2	"App-factory / lifespan + lazy torch-cv2 imports"	? ? "Belongs with the async-job slice (#16)" ? #16 is now done; unblocked	Phase-1 target (P5)
O1	"WSL temp/frame/cache + Gemini remote-file GC"	? ? "staged videos, PNG frame dumps, `*_wasbcache`, and orphaned Gemini uploads are never cleaned"	Phase-2 target. The WSL half is partially handled by `keep_frames=false`; the Gemini half is new.
D13	"`supervision` ByteTrack deprecation"	? ? pinned `<0.30.0` as stopgap; migration to `trackers` package needed	New finding in this doc (see §3b)
D14	"`tests/` reorganization to reduce fragmentation"	? ? owner's design choice on grouping metric needed first	New finding in this doc (see §3b)

Not absorbed (kept in BACKLOG as feature-gap, not code-cleanup):
- ? **Train/eval separation guardrails** ? `splits.py` exists but needs collector-side manifest change; this is a cross-repo feature, not a code-cleanup item.

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4. Design / architecture ? Phase 1 approach

4a. D3 ? Typed-config migration (first target)

What: Convert all call sites that use `config.get("indexing",{}).get(...)` to typed `AppConfig` attribute access.

Reuse map:
- `backend/config/models.py` ? `AppConfig`, `IndexingConfig`, `ValidationConfig` (single source of truth for defaults + types)
- `backend/config/__init__.py` ? re-exports `load_config`, `AppConfig`
- `backend/pipeline/validators/*.py` ? 6 validators that consume `config: Dict[str,

Any]`
- `backend/pipeline/segmenters/motion.py` ? uses `self.config.get("indexing",{})`
- `backend/reporting/__init__.py` ? already accepts both typed and dict; normalizes via `model_dump()`

Plan: Convert one module at a time, bottom-up. Validators first (they receive config as a dict, change the contract to `AppConfig`), then segmenters, then remove the `.get()` compat shim and flip to `extra='forbid'`.

Test strategy: Every module has existing tests. The config models have `test_config.py` that pins defaults. No new test behavior needed ? existing tests should pass byte-identically.

4b. B5 ? Segmenter validator fix (second target)

What: Make `segmenter_must_be_registered` actually validate against `segmenter_registry`.

File: `backend/config/models.py`, field validator on `IndexingConfig.default_segmenter`.

Risk: Near-zero. A typo'd segmenter in `config.json` that currently "works" (passes validation, fails at runtime) would now fail at startup. This is the DESIRED behavior ? catching config errors early.

4c. D2 ? App-factory refactor (third target, largest)

What: Replace `db_instance: Optional[Database] = None` + `global_config: Optional[AppConfig] = None` module globals with a `create_app(config) -> FastAPI` factory that injects state via `app.state`.

Reuse map:
- `backend/main.py` ? FastAPI app definition, all endpoints
- `backend/api/job_worker.py` ? resolves `db_instance`/`global_config` through `import backend.main as _main`
- `tests/test_api_endpoints.py` ? already uses `TestClient`; would switch to `create_app(test_config)`
- `tests/test_async_jobs.py` ? already monkeypatches `backend.main`

Plan: This is the highest-risk Phase-1 item but still a pure refactoring. The key insight: Starlette's `app.state` is the standard way to attach shared state; FastAPI's dependency injection can access it. Every endpoint that accesses `db_instance`/`global_config` would use `request.app.state.db` / `request.app.state.config` instead. The worker would receive these explicitly rather than resolving through the module object.

Test strategy: All existing tests must pass byte-identically. The `TestClient` usage changes from `from backend.main import app` to `from backend.main import create_app; app = create_app(test_config)`.

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5. Honest limits ? what Phase 1 does NOT do

- **Does NOT change any behavior.** The Gemini provider still returns `[]` on failure; the PTS validator still has a hardcoded 10.0s gap.
- **Does NOT fix B2 (Success/Failure in Gemini).** That changes the return contract callers depend on ? deferred to Phase 2.
- **Does NOT split `main.py` (D1).** The god-module split is deferred ? D2 (app-factory) is a necessary precursor.
- **Does NOT address any operational risks (O1?O5).** Those need behavioral changes.
- **A green suite after Phase 1 does NOT mean the code is "clean."** It means the structural debt targeted by D2/D3/B5 is resolved. The remaining 28 items still need attention.

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6. Deliverables & progress tracker ? **source of truth**

Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking.

ID	Deliverable	Depends on	Gated?	Status	PR
P0	**Docs-only PR ? this project doc** + index in `docs/README.md`	?	Owner approval	?	[#352](https://github.com/Khelsutra/badminton-highlight-indexer/pull/352)
?	**WASB cache test fix (Windows)** ? 2 pre-existing failures	?	No	?	[#353](https://github.com/Khelsutra/badminton-highlight-indexer/pull/353)
P4	**B5: Fix `segmenter_must_be_registered` validator**	P0	No	?	[#354](https://github.com/Khelsutra/badminton-highlight-indexer/pull/354)
P1	**D3-a: Convert validators to typed `AppConfig`** (6 files)	P0	No	?	[#355](https://github.com/Khelsutra/badminton-highlight-indexer/pull/355)
P6a	**Q3: Deduplicate `_default_db_path`**	P0	No	?	[#356](https://github.com/Khelsutra/badminton-highlight-indexer/pull/356)
P6b	**Q4: Cache builtin defaults**	P0	No	?	[#358](https://github.com/Khelsutra/badminton-highlight-indexer/pull/358)
P6c	**Q5: requirements.txt torch comment + Q6: isort (I) + Q8: cv2 comment**	P0	No	?	[#359](https://github.com/Khelsutra/badminton-highlight-indexer/pull/359) [#360](https://github.com/Khelsutra/badminton-highlight-indexer/pull/360) [#361](https://github.com/Khelsutra/badminton-highlight-indexer/pull/361)
T1	**Tier 1: Q2 (motion return type) + D11 (codec docstring)**	P0	No	?	[#362](https://github.com/Khelsutra/badminton-highlight-indexer/pull/362)
T2	**Tier 2: B6 (resolution_fps fallback + warning log)**	P0	No	?	[#363](https://github.com/Khelsutra/badminton-highlight-indexer/pull/363)
T3	**Tier 3+4: D12 (sys.path hack) + B3 (PTS gap config) + O5 (lock note) + D7 (centralize TUNED)**	P0	No	?	[#364](https://github.com/Khelsutra/badminton-highlight-indexer/pull/364)
P6d	~~D7: Centralize hardcoded tuning constants~~ ? folded into T3	P0	No	?	[#364](https://github.com/Khelsutra/badminton-highlight-indexer/pull/364)
P2	**D3-b: Convert segmenters to typed `AppConfig`** (15 files: 5 segmenters + 4 detectors + worker + eval + tools)	P1	No	?	[#366](https://github.com/Khelsutra/badminton-highlight-indexer/pull/366)
P3	**D3-c: Remove `.get()` compat shim; flip `extra='forbid'`** (18 files, +87/-169)	P2	No	?	[#367](https://github.com/Khelsutra/badminton-highlight-indexer/pull/367)
P5	**D2: App-factory refactor** ? replace module-level Optional globals with `create_app(config, db) ? FastAPI`	P3, P4	No	?	[#370](https://github.com/Khelsutra/badminton-highlight-indexer/pull/370)
P7	**Phase-1 verification:** full suite green, coverage ? 84%, mypy ratchet	P1?P6	No	?	1463/1463 pass (#371); `backend.api.job_worker` lifted from mypy quarantine
?	?	?	?	?	?
P8	**B2: Error-flagging in Gemini provider metrics** (Phase 2)	P7	No	?	[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373)
P9	**B1: Remove fabricated data from motion segmenter** ? random server/receiver/shot_count/speed ? `None`; events dropped	P7	No	?	[#385](https://github.com/Khelsutra/badminton-highlight-indexer/pull/385)
P10	**D1: Split `main.py` god-module** (~1450 lines ? smaller modules)	P7	No	?	?
P11	**O1: Gemini remote-file GC** ? per-call `_safe_delete` (#373) + startup orphan sweep scoped to our `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded	P7	No	?	[#380](https://github.com/Khelsutra/badminton-highlight-indexer/pull/380)
P12	**D13: ByteTrack deprecation** ? migrated `sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken)	P7	No	?	[#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386)
P13	**D4: Deep config merge** ? section-level replace ? recursive merge	P7	No	?	[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373)
P14	**O2: Single-executor risk** ? documented with mitigations in job_worker.py	P7	No	?	[#374](https://github.com/Khelsutra/badminton-highlight-indexer/pull/374)
P15	**Q7: Type-clean `motion.py`** ? ratchet exercise	P7	No	?	[#373](https://github.com/Khelsutra/badminton-highlight-indexer/pull/373)

| **P17** | **B2 (cont.): finish the `Success`/`Failure` return contract** ? the 4 callers that dropped provider error metrics (`trajectory_hybrid`, `yolo_hybrid`,

`gemini_refine.refine_window`, `gemini_refine.full_video_rallies`) now surface `metrics["error"]` loudly instead of silently treating a failure as "0 rallies". (Full `Success`/`Failure` return-type migration of the hybrids is follow-up.) | P8 | No | ? | [#378](https://github.com/Khelsutra/badminton-highlight-indexer/pull/378) |
| **P16+** | Phase 3 items (B4/B7/B8/D5/D6/D8-D10/D14/O3/O4) | P15 | No | ? | ? |

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7. Open questions

1. **Q: Should D1 (split `main.py`) be Phase 1 or Phase 2?** ? Leaning Phase 2: D2 (app-factory) is a natural precursor, and D1 is lower risk after the globals are gone. Owner to confirm.
2. ~~Q: Should Phase 1 include enabling `"I"` (isort) in ruff.toml?~~ ? ? Shipped in #360.

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8. Definition of done

- [x] P0: This doc approved and merged to `master`
- [x] P1?P3: Typed-config migration complete ? no `.get()` dict access remains in backend code; `extra='forbid'` on `AppConfig`
- [x] P4: `segmenter_must_be_registered` actually validates against the registry
- [x] P5: `create_app()` injects shared state via `app.state` (handlers read `request.app.state`); `backend.api.job_worker` removed from mypy quarantine. (The Optional module globals `db_instance`/`global_config` are **retained as a transitional back-compat shim** and `backend.main` stays mypy-quarantined ? full removal is follow-up, see D1/P10.)
- [x] P6: Quick wins Q1?Q6, Q8 landed
- [x] P7: Full test suite ? 1463/1463 pass (#371); `ruff` + `mypy` clean; `backend.api.job_worker` lifted from quarantine
- [x] All PRs are single-purpose, branched from `master`, with descriptive commit messages
- [ ] `docs/README.md` updated to reflect the project's terminal state
- [ ] Final status flipped to `DONE` and this doc moved to `docs/archives/past_projects/`

8.1 ? Next items ? Phase 2 pick-up (prioritized)

All items below are **Claude-doable**, small-PR-friendly, and unblocked. Pick up in order:

Immediate (HIGH impact, low risk)

#	ID	What	Why
~~1~~	~~**B2 / P17**~~	~~**Finish** the `Success`/`Failure` return contract~~ ? **? DONE (this PR).** The 3 callers that dropped provider error metrics now surface `metrics["error"]` loudly. **Follow-up remains:** migrate the hybrid segmenters' `process_video` return type from bare `List[Dict]` to the ABC's `Result[...]` (`Success`/`Failure`) ? currently suppressed with `type: ignore[override]` in `trajectory_hybrid.py` (motion.py is already fully migrated).
~~2~~	~~**O1 / P11**~~	~~Gemini remote-file GC~~ ? **? DONE (#380).** Per-call `_safe_delete` (#373) + startup orphan sweep scoped to `UPLOAD_TAG`-tagged files, grace-gated, HTTP-timeout-bounded.

Near-term (MEDIUM impact)

#	ID	What	Why
~~3~~	~~**B1 / P9**~~	~~Remove fabricated data from `motion.py`~~ ? **? DONE (#385).** Random server/receiver/shot_count/speed ? `None`; events dropped.
~~4~~	~~**D13 / P12**~~	~~ByteTrack deprecation migration~~ ? **? DONE.**

`sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken ? filed Roboflow issue). |
| 5 | **D1 / P10** | Split `main.py` god-module | ~1450 lines. D2 (app-factory) was the necessary precursor ? now done. Extract: CLI handlers ? `backend/cli/`, upload routes ? already in `backend/api/uploads.py`, debug helpers ? `scripts/`. |

Later (deferred)

#	ID	What
6+	**P16+**	B4, B7, B8, D5, D6, D8?D10, D14, O3, O4 ? Phase 3 items

> **Removed from this list ? already shipped (see the §6 tracker); they were stale entries left from before Round 3/4 landed:** D4/P13 deep config merge (#373 ? `loader.py::_deep_merge`), O2/P14 single-executor risk documented (#374 ? `job_worker.py`), Q7/P15 `motion.py` type-cleaned (#373 ? no longer in the `mypy.ini` quarantine list).

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9. References

- **Audit source:** 33 findings from a deep whole-repo read on 2026-06-24, plus 2 absorbed from `docs/BACKLOG.md` (D13, D14) for a total of 35. Two additional BACKLOG items (D2, O1) were already independently found and are cross-referenced in §3e.
- **Prior hardening sweep:** `docs/CODE_AUDIT_AND_TEST_HARDENING.md` (completed 2026-06-14) ? the previous audit-driven hardening round. This project is the next wave.
- **BACKLOG.md** `docs/BACKLOG.md` ? deferred items from the 2026-06-10 sweep; §3e maps the absorbed items.
- **CODE_MAP.md** `docs/CODE_MAP.md` ? placement rules, gotchas, and the cross-cutting footguns list.
- **mypy.ini** ? quarantine list and the ratchet instruction.
- **ruff.toml** ? lint rules; `"I"` (isort) deferred.
- **PROJECT_DOC_TEMPLATE.md** `docs/PROJECT_DOC_TEMPLATE.md` ? the template this doc follows.

Changelog
- `2026-06-24` ? Initial doc created from 33-item deep audit.
- `2026-06-24` ? Absorbed 2 findings from `docs/BACKLOG.md` (D13, D14). Cross-referenced D2 and O1 in §3e.
- `2026-06-24` ? Phase 1 begun. P0 (#352) ? #353 ? P4/B5 (#354) ? P1/D3-a (#355) ? P6a/Q3 (#356). Status ? IN PROGRESS.
- `2026-06-24` ? **Round 2.** Quick wins (#358 Q4, #359 Q5, #360 Q6 isort, #361 Q8) ? Tier 1 (#362 Q2+D11) ? Tier 2 (#363 B6) ? Tier 3+4 (#364 D12+B3+O5). 11 items shipped total.
- `2026-06-24` ? **Round 3.** P2/D3-b segmenters (#366) + P3/D3-c extra='forbid' (#367). 18 items shipped.
- `2026-06-24` ? **Round 4 (final).** P5/D2 app-factory (#370) + test fixes (#371). Status ? PHASE 1 COMPLETE. Suite 1463/1463 green. §8.1 added with prioritized Phase 2 pick-up.
- `2026-06-25` ? **§8.1 reconciled with the §6 source-of-truth (docs-only).** Removed three already-shipped items that §8.1 still listed as deferred future work ? D4/P13, O2/P14, Q7/P15 (all merged in #373/#374) ? and split the unfinished half of B2 out of the done P8 into a new **P17** (the provider return contract + the `trajectory_hybrid`/`gemini_refine` callers that still drop `metrics["error"]`). Grounded against `loader.py::_deep_merge`, `mypy.ini`, `backend/results.py`, and `backend/providers/gemini.py` + its callers. No code change.
- `2026-06-25` ? **Accuracy fixes (docs-only).** (1) §8 DoD P5 said the module globals were "replaced with `app.state`"; corrected ? `create_app()` *added* `app.state` and lifted `job_worker` from quarantine, but `db_instance`/`global_config` remain as a transitional shim and `backend.main` is still mypy-quarantined (`backend/main.py:264`, `mypy.ini`). (2) `main.py` size "~1300" ? "~1450" (now 1448 lines) in §3b/§6/§8.1.
- `2026-06-25` ? **P17 done (B2 finish).** The 4 callers that dropped provider error metrics (`trajectory_hybrid.py`, `yolo_hybrid.py`, `gemini_refine.refine_window`, `gemini_refine.full_video_rallies`) now check `metrics["error"]` and log loudly instead of silently treating a provider failure as "0 rallies" ? matching the pattern the `gemini` segmenter already uses. (`yolo_hybrid` was caught during PR #378 review ? it bound `metrics` but

never read it.) Suite 1484/1484 green; ruff + mypy clean. Follow-up: migrate the hybrids' `process_video` return type from bare `List[Dict]` to the ABC `Result[...]` contract (motion.py is already migrated; hybrids suppressed with `type: ignore[override]` for now).
- `2026-06-25` ? **P9 done (B1 finish).** Removed fabricated `random` data from the motion segmenter ? server/receiver/shot_count/avg_speed_kmh ? `None`, serve/smash/foul/winner events dropped entirely (motion was the only `add_event` caller). PR #385 review caught a real regression: `run_process_and_stitch.py` crashed on `None >= min_shot_count` (the `.get(?, 0)` default no longer applied) ? fixed with a None-guard matching the API path. Suite 1492/1492 green; ruff + mypy clean.
- `2026-06-26` ? **P12 CI fix (#386).** The D13/P12 PR failed the `full-suite` lane with `ModuleNotFoundError: No module named 'trackers'`. Two root causes: (1) the two new tests hard-imported the standalone `trackers` package, but it's a git-only dep (PyPI wheel is metadata-only ? roboflow/trackers#474) deliberately omitted from `pyproject.toml`, and the lane installs via `pip install -e .[dev]`; (2) the git ref `@v2.5.0` was wrong ? roboflow/trackers uses bare version tags (`2.5.0`), so pip failed with `pathspec 'v2.5.0' did not match`. Fixes: added a `pip install trackers @ git+?@2.5.0` step to both `ci.yml` and `nightly.yml` full-suite lanes (same out-of-band pattern as torch); fixed the tag in `requirements.txt`; and guarded the package-importing test with `pytest.importorskip("trackers")` (matching the repo's `obsreport` soft-dep convention). All 10 CI checks green.

18. user (2026-07-02T04:48:20.749Z)

Data in GCS ? the corpus source-of-truth map

> **Scope:** this map covers the **training corpus only** (`gs://khelsutra-rally-corpus`). **User-uploaded
> videos are a SEPARATE, already-shipped surface** ? `POST /api/assets` + the per-request `medium_uri` +
> `GET /api/capabilities.storage` (see `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) ? and must stay physically
> isolated from the corpus bucket (the **D1** compliance boundary: consent / no-minors / not-training holds at
> the storage layer). The deploy layer's bucket names now live in one place (`deploy/gcp/buckets.env`).

Status: LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.
Last inventoried: 2026-06-22 against `gs://khelsutra-rally-corpus`.
Goal (owner, 2026-06-22): run **all** training + inference from the cloud with **all data in GCS**,
so the local Windows box (and the intermittently-mounted external drive) is **no longer a blocker**.

> **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-corpus`, not
> from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket yet, it's in the
> **gap table (§5)** ? flag it / migrate it, don't hard-code a local path.

This doc is the **data map**. The *machine* runbooks are [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
(spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU + train/infer).
The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
(`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ? it records
where the bytes are and *what's still missing*.

---

1. The bucket

Bucket	**`gs://khelsutra-rally-corpus`** (single corpus bucket)
Project	`gen-lang-client-0306026826` (billing *Khelsutra Billing*)
Region	`asia-south1` (Mumbai), UBLA, co-located with the GPU VMs
Auth	SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com` (`roles/storage.objectAdmin`); key at `~/.gcp/rally-worker-sa.json` (local only, **never committed**). Interactive: `gcloud auth login`.
Size today	~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus)

The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gcs://` (or `gs://`),
`s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no code change**.

---

2. Canonical layout (the target contract)

The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage:
`DECOUPLED_COMPUTE/04`):

Prefix	Holds	Who reads/writes it
`videos/`	source videos / proxies (`<clip>.mp4`)	**input** to `worker infer` & `worker train`
`labels/`	golden rally labels (`<clip>.rallies.csv`)	**input** ground truth to `worker train` + all eval
`trajectories/`	cached WASB trajectory CSVs	? **eval/litmus input** (GPU-free), consumed via `manifests/golden.json`. **NOT** read by `worker train` (it synths or re-extracts)
`weights/<version>/`	trained Gen-0 bundles (`weights.json` + a run `manifest.json`)	**output** of `worker train`
`models/`	upstream pretrained weights (`wasb_badminton_best.pth.tar`)	**staged to a local path** (`$WASB_WEIGHTS` / `~/models/?`) for the detector ? not read from `gs://` in place
`outputs/<video_id>/`	per-video inference outputs (intervals, etc.)	**output** of `worker infer`
`highlights/`, `promo/`	rendered reels + promo artifacts	output / demo

> **Note on the manifest.** `worker train` does **not** read a pre-staged corpus `manifest.json` ? it
> *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories **synthetically
> by default** (`--trajectory-source synth`, CPU-mock) or for real via opt-in `--trajectory-source
> detector` (needs `videos/`). A `manifest.json` only appears as an **output** inside each
> `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable manifest
> (§7.3) ? and `manifests/golden.json` is currently consumed by **no code yet** (eval is local-FS only;
> see Known blockers).

---

3. What's actually in the bucket today (2026-06-22)

> **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical `labels/<date>_<name>.rallies.csv`
> (= original-6 + `GX010137`/`GX010141` + 7 new; the 7 old `_proxy`/bare dupes were deleted), and
> `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot; the remaining gap is **videos**
> (14 of 15) ? see §6/§7b/§8.

| Prefix | Present | Drift / notes |

|---|---|---|
`labels/`	7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy`	? **inconsistent naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`, `GX010141`
`videos/`	3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`	? corpus videos/proxies almost entirely absent
`trajectories/`	?	? **does not exist** (every trajectory is local-only)
`models/`	`wasb_badminton_best.pth.tar`	? (? WASB-C3 weight-provenance still an open legal item before any external share)
`weights/`	`0.1.0-l4/`	? first real L4 train bundle
`outputs/`	one `<video_id>/`, `parity/`	? dry-run outputs
`highlights/`, `promo/`	reels + promo	?

Drift summary: the bucket has extra prefixes (`highlights/`, `promo/`, `models/` alongside `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the corpus inputs** (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a complete corpus.

---

4. How the cloud consumes it

Provisioning + the verified commands live in [`deploy/gcp/README.md`](../deploy/gcp/README.md) §3?§5.
The data-facing entry points (`backend/worker/__main__.py`):

```bash
# config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
#   { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826" } } }

# INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
python -m backend.worker infer \
  --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
  --out-uri   gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json

# TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds manifest on-box,
# runs training.gen0.harness, pushes weights/<version>/):
RALLY_DETECTOR_IMPL=native python -m backend.worker train \
  --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
  --version <v> --trajectory-source detector --segmenter wasb_hybrid --config config.json
```

A bare/relative path resolves under a configured `input_root`/`input_backend` (`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-relative.

---

5. The gap ? what's still local-only (why the box is still a blocker)

All of this is **tiny except the originals** ? the metadata that blocks cloud eval is < 25 MB total:

Data	Where it is now	Size	In bucket?	Needed for
20 WASB trajectory CSVs	`C:\?\Annotation Setup\Collect\Trajectories\`	**17.5 MB**	? none	eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-extracts.)*
golden manifest	`output/human_lovo_manifest.json` (C:-absolute paths)	4.5 KB	?

eval corpus definition |
original-6 labels	`F:\[Khelsutra]?\Training\Golden Labelled\*.rallies.csv`	~12 KB	?	train + eval
7 new-clip labels	repo `output/*.rallies.csv` (gitignored) **+** F: archive (backed up 2026-06-22)	~12 KB	? partial/inconsistent	train + eval
source videos / proxies	F: archive + Google Drive golden folder	**100s of GB**	? 3 only	infer + train (real WASB extraction)

> The corpus *metadata* (trajectories + labels + manifest, ~25 MB) is what actually blocks GPU-free
> cloud eval and the served Gate-A litmus ? and it's trivially small to migrate. The **videos** are the
> only large migration, and they already exist on **Google Drive** (the golden-corpus folder), so they
> can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path entirely.

---

6. Migration plan (close the gap)

Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker). Stage all trajectories,
all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`. Pending the §7
naming decision. Sketch:

```bash
# trajectories (decision-free; absent today)
gcloud storage cp "C:\?\Collect\Trajectories\*.csv" gs://khelsutra-rally-corpus/trajectories/
# labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv
# eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)
```

Phase 2 ? videos ? `videos/<name>.mp4` (bucket name = the eval `name`, consistent with
`trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see `scratch/copy_videos_to_gcs.ps1`):

Source	Clips	Size
Drive golden folder `?/Golden Label Videos Request - June 15/[Done] Video N/` (**proxies**)	`mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles` (7)	~13.5 GB
F: archive (**full MP4s** ? proxy-first before upload)	`GX010137`, `GX010141`, + original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`, `testlarge_short`, `mahadevpura_singles`) (8)	~50 GB

Two ways to run it:
- **Decoupled (preferred ? no local box):** `rclone` on a cloud VM with a Google-Drive remote ?
  `rclone copy gdrive:"<?/[Done] Video N>/<proxy>.mp4" :gcs:khelsutra-rally-corpus/videos/<name>.mp4`
  (one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>` on the box for
  the shared folder. This is the path that actually removes the local box.
- **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh scratch/copy_videos_to_gcs.ps1
  -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses local bandwidth.

Priority: the 7 Drive **proxies** first (already 1080p, sufficient for detection). The 8 large F:
full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode)

before upload.
? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`, `videos/mahadevpura-1.MP4`)
are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-dupe-then-delete
(same as the label cleanup). Track per-clip in §8.

Phase 3 ? make eval/litmus read GCS (real code, NOT a config flip). The eval drivers
(`served_gate_a.evaluate_video`, `calibrate_wasb.load_golden_gts`, `TrackNetRunner.parse_trajectory_csv`)
open `spec["trajectory"]`/`spec["labels"]` with plain `open()` and gate on `os.path.isfile()` ? they
have **no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3 must thread
`storage.fetch`/`resolve_input_uri` through those paths first. Then: re-probe `fps`/`frame_width` from
the bucket videos (don't trust the copied values ? a wrong fps silently mis-aligns every label), and
re-measure the served Gate-A litmus on the GCS 15-clip set as a [[launch-report-convention]] event
(dual-set RAW+golden no-regression, not a silent eval-set swap ? see PR #236).

---

7. Decisions

1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-DD>_<name>.rallies.csv`, where
   `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring date** =
   the *earliest* mtime across all known copies (C: working copy + F: archive ? robust to copy-resets;
   a restored C: copy shows today, the F: archive keeps the real date). Fixes the `_proxy` inconsistency
   and disambiguates recycled GoPro names by date. *(Future, when videos are in the bucket: optionally
   add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each video.)*
2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so cloud/CI eval +
   the served Gate-A litmus run **GPU-free** (training can still regenerate via `--trajectory-source detector`).
3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-corpus/manifests/golden.json`
   (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-path manifest;
   flipping the litmus to read the GCS manifest is Phase 3.
4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the 15-clip golden
   set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the local box).

> Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp the label's
> authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in gitignored
> `scratch/`; promote it to `scripts/` if corpus-sync recurs).
> ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects (`*_proxy.rallies.csv`,
> bare `GX010128`/`mahadevpura-1`) were confirmed **byte-identical (MD5)** to the new canonical names
> (content also on C:/F:) and **deleted** ? `labels/` now holds exactly the 15 canonical objects.

> ? **gcloud gotcha (Phase-2):** `gcloud storage cp` treats `[...]` as a glob, so source

paths under
> `[Done] Video N/` (Drive) or `[Khelsutra] ?` (F:) fail with "matched no objects." The owner-box
> script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through a bracket-free temp file; the
> `rclone` cloud path has no such issue.

7b. Known blockers ? must-fix before the DoD (2026-06-22 adversarial review)

1. **? RESOLVED 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319))
   ? dated labels broke `worker train --trajectory-source detector`.** *History:* `cmd_train` derived the
   join key as `fname.replace(".rallies.csv","")` ? `2026-06-22_GX020094`, then looked up
   `video_by_stem.get(name)` against bare video stems (`GX020094`), so every clip was skipped and train
   aborted with `<2 videos`. **Fix:** the label?clip-name derivation is now the helper `_label_clip_name()`
   in [backend/worker/__main__.py](../backend/worker/__main__.py), which strips both the
   `.rallies.csv`/`.csv` suffix AND a leading `^\d{4}-\d{2}-\d{2}_` prefix, so a dated label and a legacy
   bare label both resolve to the bare `videos/<name>` stem (preserves the locked human-readable naming).
   Tested: parametrized helper cases + a dated-label?bare-video `cmd_train` join regression + per-label
   skip-branch coverage. *(`synth` train + all eval were unaffected ? they don't join videos.)*
2. **Phase-3 eval-on-GCS is net-new plumbing, not a flag** ? see §6 Phase-3. Eval is local-FS only.
3. **`manifests/golden.json` is consumed by zero code today** (train ignores it; eval can't fetch
   `gs://`) ? it's correct-but-dormant until blocker #2 lands. Add a tiny manifest-validator + a CI job
   that authenticates to GCS and runs the litmus against it, or the DoD ("CI reads only from `gs://`")
   stays unverifiable.
4. **Gating infra (owner-only, not done):** GPU `GPUS_ALL_REGIONS` quota is still **0** (no VM ever
   created ? `deploy/gcp/README` §2); cloud-box secrets (SA JSON + rotated `GEMINI_API_KEY` + WASB
   weights staged from `models/`) aren't set up persistently. No real cloud train/infer runs until these.
5. **Reproducibility:** the migration + copy helpers live in gitignored `scratch/`
   (`migrate_golden_to_gcs.py`, `copy_videos_to_gcs.ps1`) ? promote to `scripts/` so Phase-2/re-syncs
   are repeatable from the repo, and add a referenced **proxy-generation** step for the 8 large F: originals.

8. Progress tracker

Item	Status
Bucket + SA + provisioning (deploy/gcp)	? built
Cloud worker `infer`/`train` by URI	? built (`backend/worker`)
This data-map doc	? 2026-06-22
Phase 1: trajectories ? `trajectories/` (15)	? 2026-06-22
Phase 1: labels reconciled ? `labels/<date>_<name>` (15)	? 2026-06-22
Phase 1: eval manifest ? `manifests/golden.json`	? 2026-06-22
Phase 1: remove 7 stale `_proxy`/bare label objects	? 2026-06-22 (MD5-verified dupes, deleted)
Blocker §7b#1: dated labels join in `worker train --trajectory-source detector`	? 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319)) ?

```
`_label_clip_name` strips the ISO-date prefix |
246     | Phase 2: corpus videos/proxies ? `videos/<name>.mp4` | ? mapping + script done;
**1/15** demo'd (`mahadevpura_1`); 14 remain ? run `rclone` on a cloud box (preferred) or the
owner-box script |
247     | Phase 2: proxy the 8 large F: full-MP4 clips before upload | ? |
248     | Phase 3: eval/litmus default to the GCS manifest | ? |
249
250     **Definition of done:** a fresh GPU box (or CI) can run inference, the eval suite, and
Gen-0 training
251     reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path
required.
252
253     ---
254
255     ## Related
256     - [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in
§7).
257     - [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer
(§5).
258     - [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) ·
[`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
259     - `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.
260
```

19. assistant (2026-07-02T04:48:47.189Z)

Now let me examine the Python files with potentially stale ByteTrack docstrings, and verify code facts.

20. user (2026-07-02T04:48:49.628Z)

```
1       """Person-detection rally cue ? torchvision detector + ByteTrack + motion windowing.
2
3       Extracted verbatim from `yolo_hybrid` (fusion P1, `docs/MULTI_SIGNAL_FUSION_PLAN.md`):
4       the **person cue is a swappable producer**, not logic baked into one segmenter. Today
5       `HybridYoloSegmenter` is the only consumer (it delegates Stage-1 here, byte-
identically);
6       next the fusion segmenter consumes the SAME producer over shuttle-seeded windows, and
7       later the learned classifier reads its persisted features. Keeping detection here (not
in
8       the segmenter) is what makes "add a cue = register a producer" true.
9
10      Behaviour is unchanged from the in-lined version ? only the home moved. The one
11      exception is the ByteTrack swap (D13/P12): ``make_tracker`` now builds a
12      ``trackers.ByteTrackTracker`` with its constructor params pinned to ``sv.ByteTrack``'s
13      behaviour (see that method's docstring), so the person-cue recall is preserved.
14        - `lazy_load_model`              ? load a permissively-licensed torchvision detector
15        - `make_tracker`                 ? a fresh ByteTrack tracker per chunk (standalone
`trackers` pkg)
16        - `detect_and_track`             ? one frame ? confident persons with persistent IDs
17        - `extract_action_windows_from_chunk` ? a chunk ? high-motion candidate windows
18          with the motion stats (`peak/mean_velocity`, `active_frame_count`, `track_id_count`)
19          that feed the candidate `stats` JSON and the learned fusion features.
20
21      The model libs (torch, torchvision, supervision) are imported lazily inside the methods,
and the
22      detector weights load only on first detection (`lazy_load_model`) ? so importing the
segmenter
23      registry never loads a detector model NOR requires torch (the LIGHT tier ships no torch
? P2e;
24      the person cue is HEAVY-only and never instantiated under Gemini/motion/cpu-mock), and
tests can
25      mock the seams. (cv2 stays top-level ? opencv-python is a hard dependency, always
installed.)
26      Licensing: torchvision detectors are BSD + COCO weights; ByteTrack is MIT (via the
```

```
standalone ``trackers`` package, Apache-2.0, since D13/P12; was ``supervision.ByteTrack``)
(ADR-005; ultralytics' AGPL YOLO was dropped).
"""

import logging
from typing import Any, Callable, Dict, List, Optional, Tuple

import cv2

from backend.utils.geometry import get_velocity_normalised

logger = logging.getLogger(__name__)

# torchvision detection models are trained on COCO where the "person" category is
# class 1 (index 0 is the implicit background). NB: ultralytics YOLO used class 0 ?
# this off-by-one is the classic gotcha when swapping detectors. Keep it named.
_PERSON_CLASS_ID = 1

# Permissively-licensed torchvision detectors selectable via config
# `indexing.detector_model`. All are BSD-licensed (code + COCO weights).
_DETECTOR_FACTORIES = {
    "fasterrcnn_resnet50_fpn": "fasterrcnn_resnet50_fpn",
    "fasterrcnn_mobilenet_v3_large_fpn": "fasterrcnn_mobilenet_v3_large_fpn",
    "retinanet_resnet50_fpn": "retinanet_resnet50_fpn",
}
_DEFAULT_DETECTOR = "fasterrcnn_resnet50_fpn"


class PersonDetector:
    """Stage-1 person detection + tracking + motion-window extraction (a rally cue).

    Owns the torchvision model + per-chunk ByteTrack tracker. Constructed with the run
    ``config`` (reads ``indexing.{use_gpu, detector_model, detector_score_threshold,
    min_action_window_duration}``); the model loads lazily on first use.
    """

    def __init__(self, config: Dict[str, Any]):
        self.config = config
        # Dynamic torchvision module (None until lazy_load_model); typed Any so the
        # `self.model([tensor])` inference call type-checks without a quarantine.
        self.model: Any = None
        self.device: Optional[str] = None

    def lazy_load_model(self) -> None:
        if self.model is not None:
            return

        # Imported lazily so the module imports without torch/torchvision present (the LIGHT tier
        # ships no torch ? P2e) and so the test suite (which pre-sets self.model) never pulls weights.
        import torch
        from torchvision.models import detection as tv_detection

        idx_cfg = self.config.get("indexing", {})
        if isinstance(idx_cfg, dict):
            from backend.config.models import IndexingConfig

            idx_cfg = IndexingConfig(**idx_cfg)
        use_gpu = idx_cfg.use_gpu
        self.device = "cuda" if use_gpu and torch.cuda.is_available() else "cpu"
        if use_gpu and not torch.cuda.is_available():
            logger.warning(
                "use_gpu is True but CUDA is not available. Falling back to CPU."
```

```
            )

        model_name = idx_cfg.detector_model
        if model_name not in _DETECTOR_FACTORIES:
            logger.warning(
                f"Unknown detector_model '{model_name}'. Falling back to {_DEFAULT_DETECTOR}."
            )
            model_name = _DEFAULT_DETECTOR

        builder = getattr(tv_detection, _DETECTOR_FACTORIES[model_name])
        logger.info(f"Loading torchvision detector '{model_name}' on {self.device}...")
        # weights="DEFAULT" pulls the best available COCO-pretrained checkpoint.
        self.model = builder(weights="DEFAULT")
        self.model.eval()
        self.model.to(self.device)

    def make_tracker(self, fps: float) -> Any:
        """Create a fresh ByteTrack tracker for one chunk (track IDs are chunk-local).

        D13/P12: migrated from ``supervision.ByteTrack`` (deprecated, removed in 0.30.0)
        to the standalone ``trackers`` package (``ByteTrackTracker``). Imported lazily so
        tests can mock this method without the package installed.

        The constructor params are pinned to ``sv.ByteTrack``'s behaviour so the migration
        is behaviour-preserving. The new package's defaults differ materially ?
        ``track_activation_threshold`` 0.7 (vs sv's ~0.35 effective floor),
        ``high_conf_det_threshold`` 0.6 (vs sv's ~0.35), and
        ``minimum_consecutive_frames`` 2 (vs sv's 1). Left at the new defaults, a person
        detected at confidence [0.5, 0.7) (common during fast smashes / partial occlusion)
        would never spawn a track, silently regressing rally recall; and the first sampled
        frame of every track would be emitted as -1 (dropped). Pinning all three keeps the
        0.5 ``detector_score_threshold`` pre-filter the effective floor and yields an ID on
        the first matched frame. ``minimum_iou_threshold`` is left at the package default
        (0.1), which is roughly equivalent to sv's matching gate.
        """
        from trackers import ByteTrackTracker

        frame_rate = max(1, int(round(fps)))
        return ByteTrackTracker(
            frame_rate=frame_rate,
            track_activation_threshold=0.25,  # sv default; <= detector_score_threshold (0.5)
            high_conf_det_threshold=0.35,  # match sv's det_thresh so [0.5,0.7) can spawn
            minimum_consecutive_frames=1,  # sv default: emit a real ID on the first frame
        )

    def detect_and_track(
        self, frame: Any, tracker: Any, score_thresh: float
    ) -> List[Tuple[int, Tuple[float, float, float, float]]]:
        """Detect persons in a frame and assign persistent track IDs.

        Returns a list of (track_id, bbox) where bbox is (x1, y1, x2, y2) in pixels.
        This replaces the old ultralytics ``model.track(...)`` call, which bundled
```

21. user (2026-07-02T04:48:52.101Z)

```
1:"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
2-(Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).
3-
4-Stage-1 detection now lives in `backend/pipeline/detectors/person_detector.py`
--
10-service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
11-detection uses torchvision's permissively-licensed (BSD) detectors and the
12-association/tracking that YOLO.track() used to bundle in is provided by
13:supervision's MIT-licensed ByteTrack.
14-
15-The registry key stays "yolo_hybrid" for back-compat with existing configs and the
16-DB `detector` tags, even though no YOLO is involved anymore.
--
54-    @property
55-    def description(self) -> str:
56-        return (
57:            "Two-stage pipeline: torchvision person detection + ByteTrack for fast "
58-            "candidate filtering, then an AI provider for high-precision semantic
boundaries."
59-        )
60-
--
99-            return None
100-
101-        logger.info(
102:            f"Initializing Hybrid Segmenter (torchvision + ByteTrack) for video:
{video_path}"
103-        )
104-        return provider
105-
```

22. user (2026-07-02T04:49:13.054Z)

```
2198 backend/main.py
torch>=2.12.0
torchvision>=0.27.0
supervision>=0.21.0
```

D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see + deploy/cloudrun/Dockerfile ? edit ALL of them in lockstep on a bump, or the lanes/b silently resolve different versions.

```
trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
scipy>=1.10

--- create_app/lifespan ---
90:app = FastAPI(title="Sports Highlight Indexer API")
93:def create_app(config: AppConfig, db: Database) -> FastAPI:
2184:        create_app(global_config, db_instance)
--- gemini GC ---
59:    def cleanup_orphaned_files(self, grace_sec: int = 3600) -> int:
--- M11/M12 files ---
backend/flywheel.py
backend/storage/gdrive_api.py
--- cv2 import ---
34:import cv2
```

23. user (2026-07-02T04:49:15.254Z)

```
69c6266 docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan (design)
(#389)
2028f98 docs(nightly): add person-cue Stage-1 stats lane to the engine-regression plan (design)
3624a8d feat(D13/P12): migrate ByteTrack to standalone trackers package (#386)
```

```
1283e18 docs(project): Decision-Snapshot Regression plan ? a deterministic quality-neutral proof
(#383)
cd5b3d2 feat(O1/P11): Gemini remote-file GC ? startup orphan sweep (#380)
947a36b docs: update CODE_CLEANUP_AUDIT ? Phase 1 complete, Phase 2 prioritized (#372)
9cd7598 fix(tests): resolve remaining test signature mismatches from app-factory refactor (#371)
de09ec8 refactor(main): app-factory + typed-config migration (D2/D3/P2/P3/P5/P7) (#370)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF +
consent faked-deny) (#336)
aac7633 feat(serving): M9-auth ? Cloudflare Access edge-auth + per-tenant owner_id (default-OFF)
(#290)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF)
(#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked)
(#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity
test (default-OFF) (#286)
--- gdrive_api PR ---
b6214e5 chore: ruff format the whole repo (259 files) (#387)
f523d3a refactor: enable isort (I) in ruff, auto-fix 143 files (Q6) (#360)
819e395 feat(serving): M12 ? gdrive-api:// OAuth/API StorageBackend (D14 Northstar) (#338)
--- flywheel PR ---
b6214e5 chore: ruff format the whole repo (259 files) (#387)
334ad0f feat(serving): M11 data-flywheel skeleton ? capture?shadow-eval?promote (default-OFF +
consent faked-deny) (#336)
```

24. user (2026-07-02T04:49:29.211Z)

{"mergeStateStatus":"DIRTY","mergeable":"CONFLICTING","state":"OPEN","title":"refactor(D1/P10): extract orchestration.py from main.py (god-module split, step 1)","updatedAt":"2026-06-26T14:37:18Z"}

```
Archive
Golden Labelled
Trajectories
golden_labels_badminton.csv
rally_review.csv
ls: cannot access 'C:/Users/avidu/Projects/Annotation Setup': No such file or directory
```

25. user (2026-07-02T04:49:30.939Z)

```
126:[Omitted long matching line]
127:[Omitted long matching line]
```

26. user (2026-07-02T04:49:38.560Z)

```
110     ## 7. Deliverables & progress tracker  ? **source of truth**
111     Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off
`origin/master`, no stacking. Default-OFF where it touches core callers.
112
113     | ID | Deliverable | Depends | Gated? | Status | PR |
114     |----|-------------|---------|--------|--------|----|
115     | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? |
**? APPROVED 2026-06-21** |
116     | M1 | `deployment.profile` + `compute_target` config + preset bundles +
precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117     | M2 | `input_backend`/`input_root`; wire main CLI/API input through
`StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118     | M3 | Configurable output/scratch base ? kill hardcoded `output/`
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119     | M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 |
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
```

indexer/pull/283) |
120 | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed **runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? | [#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) + **containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner (GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287) (shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) | M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds, wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation + `/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`; DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | [#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT verify + owner_id tag, default-OFF ?; **+ exposure-safety boot posture** in `startup.py` ? edge-auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see changelog; consent cols + ToS/DPA ? owner/counsel) |
125 | M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit owner approval | ? | ? |
126 | **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | **skeleton landed** (see changelog): `backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook, **consent faked-deny**; live activation ? owner/counsel (consent schema) + owned-model/golden data |
127 | **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the source from + writes the stitched reel **back to the user's Google Drive next to the source** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | **backend landed** (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` + `gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app ? owner** |
128
129 **Testing strategy is a first-class deliverable** ? [`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ? [`runlogs/`](runlogs/).
130
131 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132 **? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:**
133 1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
134 3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
135 4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to proxy).
136 5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data; counsel-gated to M9).
137 6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until the owned model clears the floor; capture?store?eval?goldenize meanwhile).
138
139 **? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:**
140 2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
141 7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ? distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
142 8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only** (speed, no output-byte change); NVENC-encode deferred behind a config knob.

```
9. ? **Input cap ? D16:** **hard-reject `<2GB`** for v1, as a config cap (raise to 4GB later in one line).
10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.
```

27. user (2026-07-02T04:51:05.592Z)

```
"""Two-stage rally segmenter: torchvision person detection + ByteTrack tracking
(Stage 1, local) feeding an AI provider for semantic boundaries (Stage 2).

Stage-1 detection now lives in `backend/pipeline/detectors/person_detector.py`
(`PersonDetector`, a reusable rally-cue producer ? fusion P1) so the SAME person cue
can be consumed by the future fusion segmenter; this segmenter just orchestrates
chunking, the AI-provider handoff, and DB population around it. Behaviour is unchanged.

Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted
service free of copyleft obligations (see docs/MONETIZATION_AUDIT.md and ADR-005),
detection uses torchvision's permissively-licensed (BSD) detectors and the
association/tracking that YOLO.track() used to bundle in is provided by
supervision's MIT-licensed ByteTrack.

The registry key stays "yolo_hybrid" for back-compat with existing configs and the
DB `detector` tags, even though no YOLO is involved anymore.
"""

import concurrent.futures
import logging
import os
import time
from typing import Any, Dict, List, Tuple

from backend.pipeline.detectors.person_detector import PersonDetector
from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
from backend.providers import provider_registry
from backend.sports import sport_registry
from backend.utils.paths import output_base
from backend.utils.video import extract_video_chunk, get_video_duration
```

28. user (2026-07-02T04:51:07.898Z)

```
tools/cleanup_caches.py
b6214e5 chore: ruff format the whole repo (259 files) (#387)
3b36f9f docs(CLAUDE.md): truth-sync the stale Current-state section to shipped reality (audit R2-5)
f28af73 P2: engine free-space guard (HTTP 507 + WSL frame-stage BLOCK) (#407)
4a7bfbe docs: record the launch NO-GO decision (owner, 2026-06-30) (#404)
```

29. user (2026-07-02T04:51:20.059Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/BACKLOG.md has been updated successfully. (file state is current in your context ? no need to Read it back)

30. user (2026-07-02T04:51:33.471Z)

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31. user (2026-07-02T04:51:45.447Z)

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32. user (2026-07-02T04:51:57.366Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

33. user (2026-07-02T04:52:02.692Z)

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34. user (2026-07-02T04:52:18.202Z)

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35. user (2026-07-02T04:52:19.488Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DOC_STATUS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

36. assistant (2026-07-02T04:52:28.606Z)

Now docs/README.md lines 41-42.

37. user (2026-07-02T04:52:35.050Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

38. user (2026-07-02T04:52:38.216Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

39. user (2026-07-02T04:53:09.235Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/NEXT_STEPS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

40. user (2026-07-02T04:53:21.791Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

41. user (2026-07-02T04:53:25.165Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

42. user (2026-07-02T04:53:34.330Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

43. user (2026-07-02T04:53:38.807Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

44. user (2026-07-02T04:53:51.046Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/CODE_CLEANUP_AUDIT_2026-06-24.md has been updated successfully. (file state is current in your context ? no need to Read it back)

45. assistant (2026-07-02T04:53:56.426Z)

Now the stale ByteTrack docstring in yolo_hybrid.py (doc-comment only).

46. user (2026-07-02T04:54:01.815Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/backend/pipeline/segmenters/yolo_hybrid.py has been updated successfully. (file state is current in your context ? no need to Read it back)

47. assistant (2026-07-02T04:54:29.644Z)

Now the DATA_IN_GCS.md §5 gap-table reconcile against its own §7/§8.

48. user (2026-07-02T04:54:32.798Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

49. user (2026-07-02T04:54:57.219Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

50. user (2026-07-02T04:55:05.405Z)

The file C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/c1fcb4aa-537d-41fd-858e-53f93349bf5a/scratchpad/wt-docs-reconcile/docs/DATA_IN_GCS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

51. user (2026-07-02T04:55:17.338Z)

```
[check-md-links] clean ? no broken internal links in live docs.
LINKCHECK_EXIT=0
```

52. user (2026-07-02T04:55:17.886Z)

```
All checks passed!
test_fusion_hybrid.py
test_tracknet_hybrid.py
test_trajectory_hybrid_equivalence.py
test_yolo_hybrid.py
```

53. user (2026-07-02T04:56:08.708Z)

```
........                                                                 [100%]
8 passed in 2.40s
IMPORT_OK
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 1 source file
```

54. assistant (2026-07-02T04:56:13.829Z)

All green. Reviewing the final diff before committing.

55. user (2026-07-02T04:56:17.365Z)

```
 M backend/pipeline/segmenters/yolo_hybrid.py
 M docs/BACKLOG.md
 M docs/CODE_CLEANUP_AUDIT_2026-06-24.md
 M docs/DATA_IN_GCS.md
 M docs/DOC_STATUS.md
 M docs/NEXT_STEPS.md
 M docs/README.md
 backend/pipeline/segmenters/yolo_hybrid.py |  6 +++--
 docs/BACKLOG.md                            |  8 +++----
 docs/CODE_CLEANUP_AUDIT_2026-06-24.md      | 11 ++++-----
 docs/DATA_IN_GCS.md                        | 36 +++++++++++++++++-------------
 docs/DOC_STATUS.md                         |  9 ++++----
 docs/NEXT_STEPS.md                         |  8 +++----
 docs/README.md                             |  4 ++--
 7 files changed, 46 insertions(+), 36 deletions(-)
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
```

56. user (2026-07-02T04:56:26.012Z)

```
1
8:> **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not**
cleared for an external/commercial launch; we stay in **development**. This is the expected
default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate
target (#172)** ? currently unset, needs corpus growth (and the R2 tolF1 ablation, item 8);
**(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing).
P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ?
launching. Re-decide only when both unblockers clear.
```

57. user (2026-07-02T04:56:36.591Z)

```
 > **? LAUNCH DECI
```

58. assistant (2026-07-02T04:56:47.487Z)

NO-GO block confirmed as untouched context. Committing with explicit paths.

59. user (2026-07-02T04:57:02.458Z)

```
docs/reconcile-trackers-2026-07-02
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
[docs/reconcile-trackers-2026-07-02 dfdd2d9] docs: reconcile stale trackers/blurbs with shipped
state (audit R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
 7 files changed, 46 insertions(+), 36 deletions(-)
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
```

60. user (2026-07-02T04:57:15.464Z)

```
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
remote:
remote: Create a pull request for 'docs/reconcile-trackers-2026-07-02' on GitHub by visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/docs/reconcile-
trackers-2026-07-02
remote:
branch 'docs/reconcile-trackers-2026-07-02' set up to track 'origin/docs/reconcile-
trackers-2026-07-02'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      docs/reconcile-trackers-2026-07-02 -> docs/reconcile-trackers-2026-07-02
```

61. assistant (2026-07-02T04:57:22.839Z)

```
Branch verified clean (only my commit) and pushed. Opening the PR.
```

62. user (2026-07-02T04:57:53.427Z)

```
https://github.com/Khelsutra/badminton-highlight-indexer/pull/420
```

63. assistant (2026-07-02T04:58:02.720Z)

```
PR opened. Final state check and summary.
```

64. user (2026-07-02T04:58:05.551Z)

docs/reconcile-trackers-2026-07-02...origin/docs/reconcile-trackers-2026-07-02

```
dfdd2d9 docs: reconcile stale trackers/blurbs with shipped state (audit
R1-13/R1-14/R1-01-doc/DATA_IN_GCS)
```

65. user (2026-07-02T04:58:36.050Z)

```
Structured output provided successfully
```